

Multivariate reconciliation for hierarchical time series

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Figures

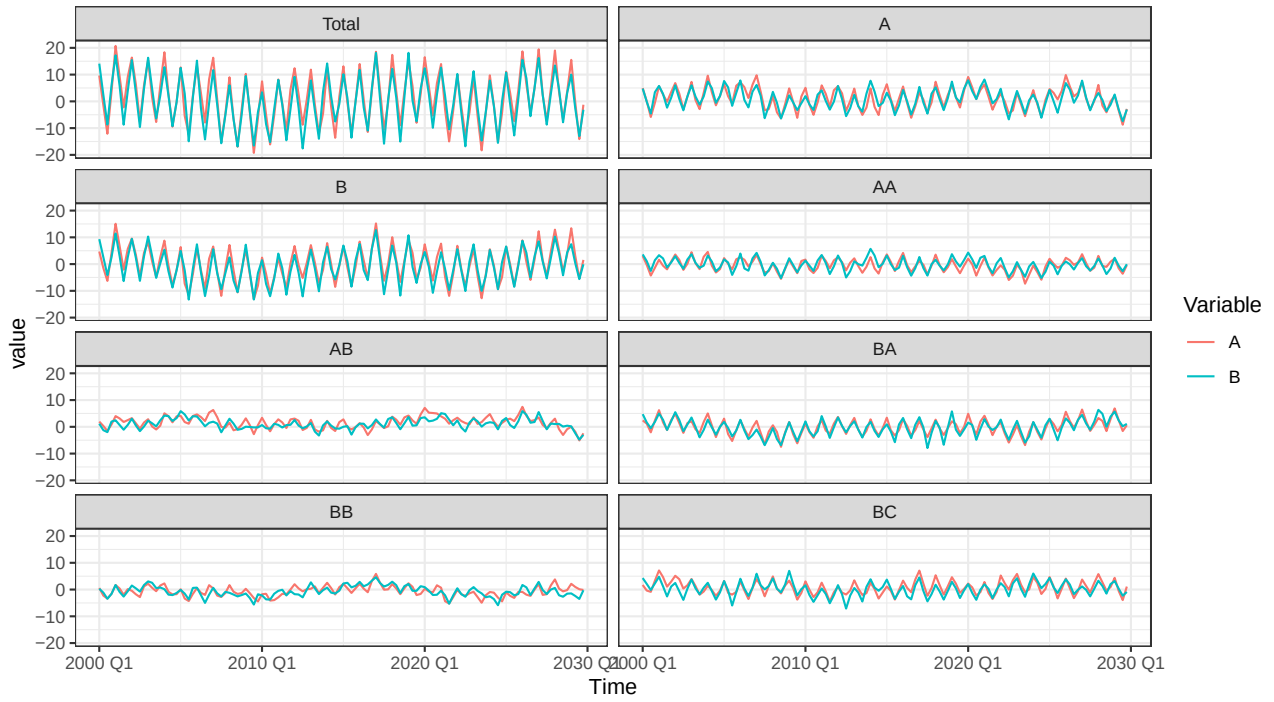


Figure S1: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 1.

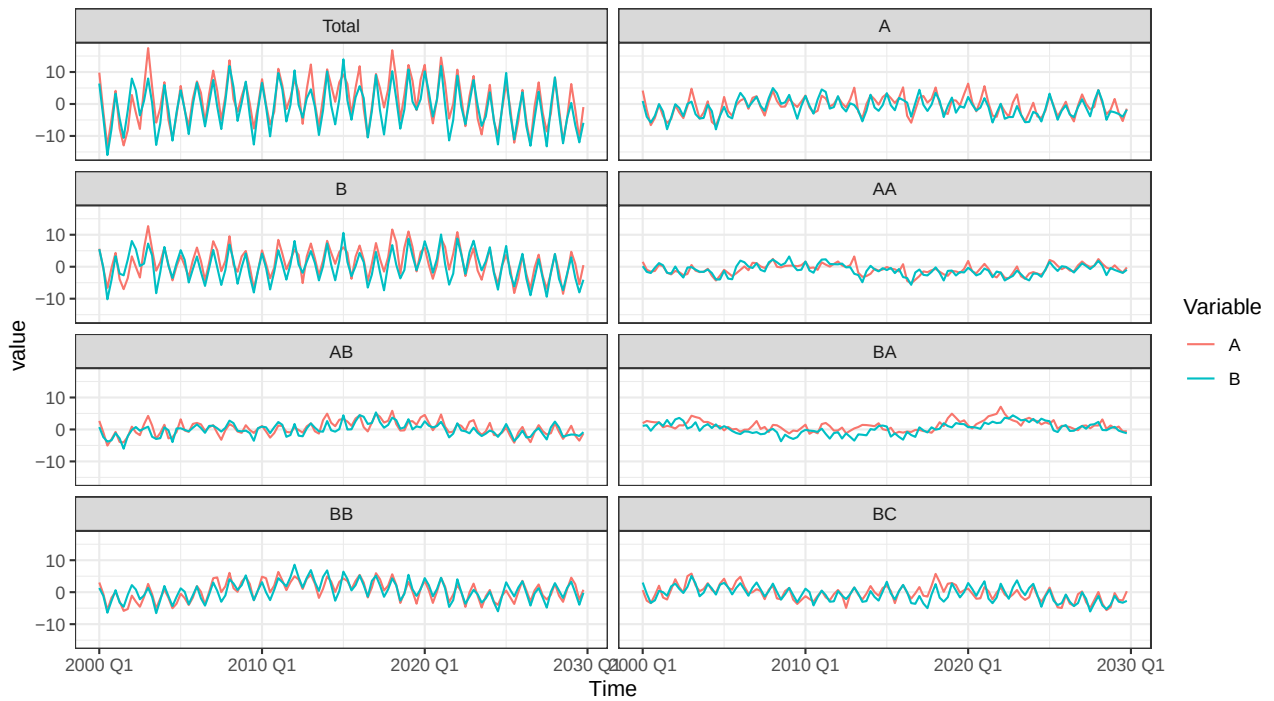


Figure S2: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 2.

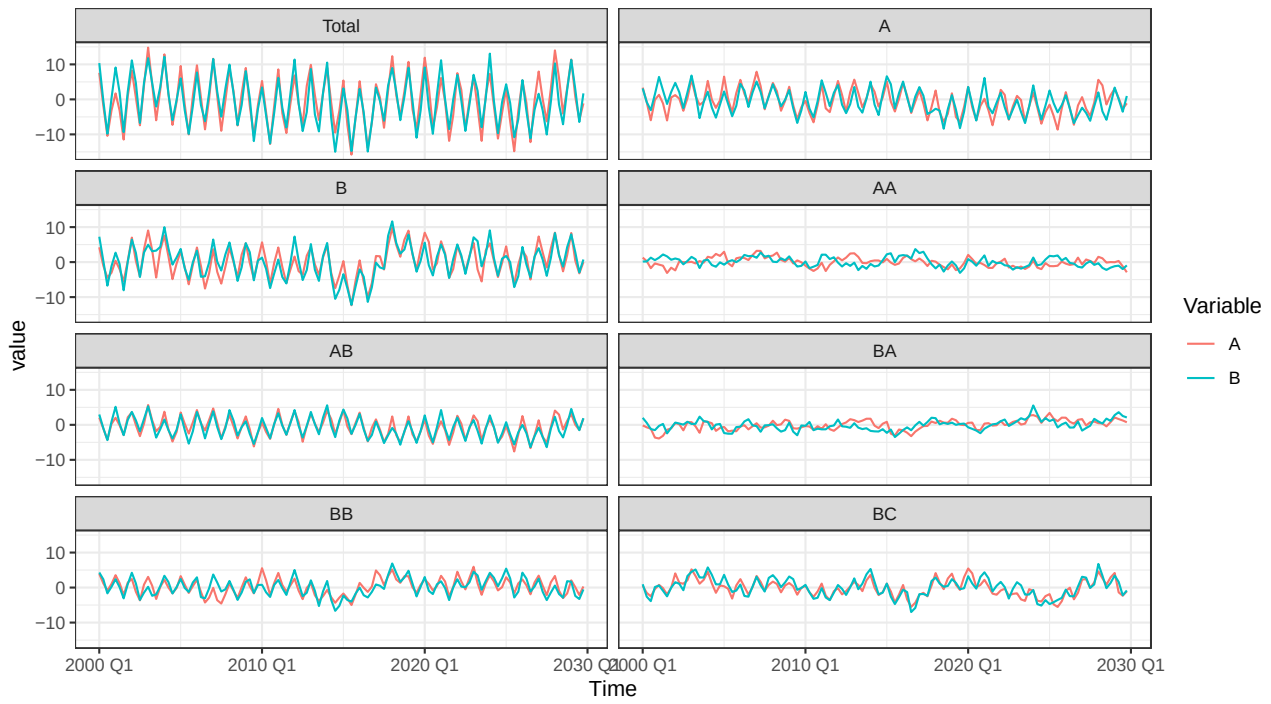


Figure S3: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 3.

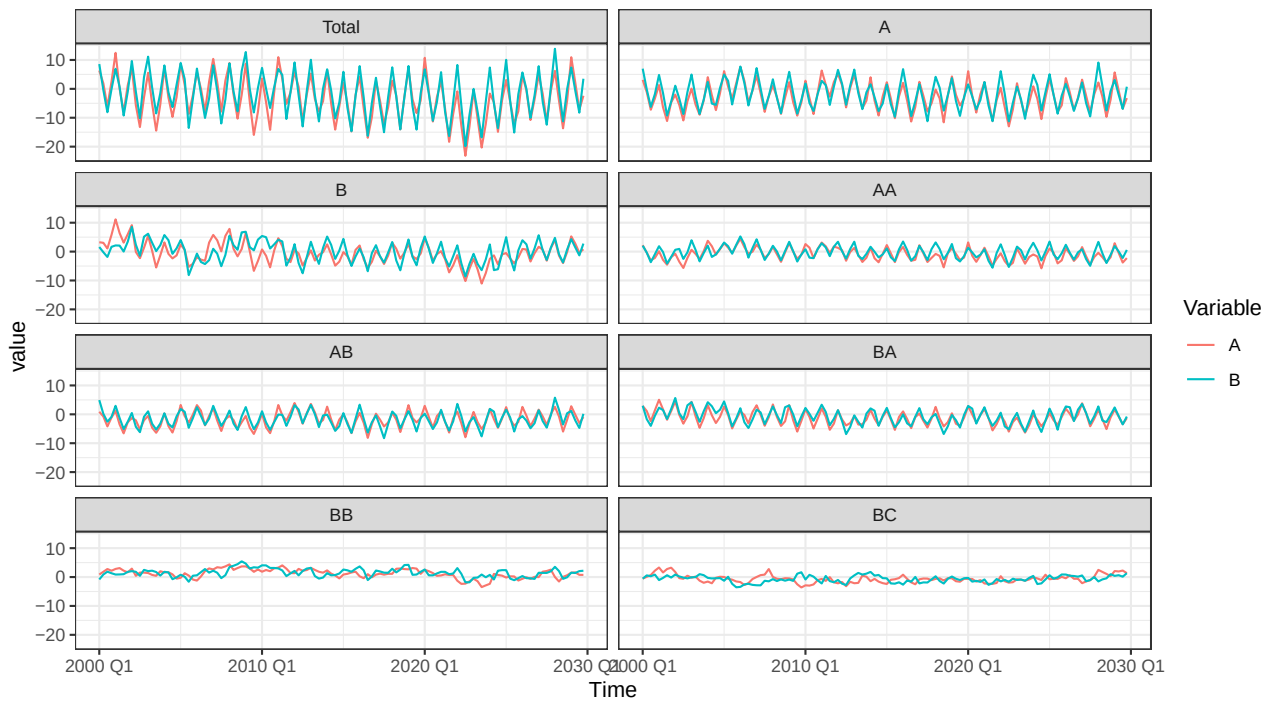


Figure S4: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 4.

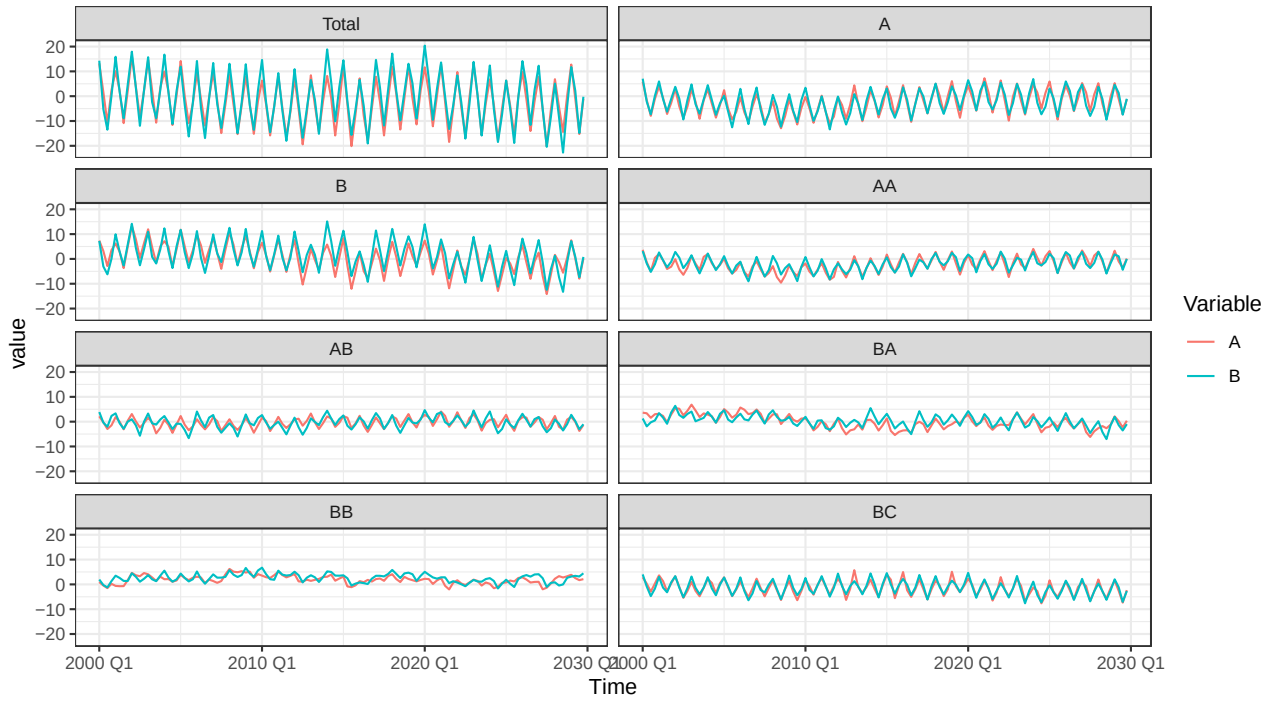


Figure S5: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 5.

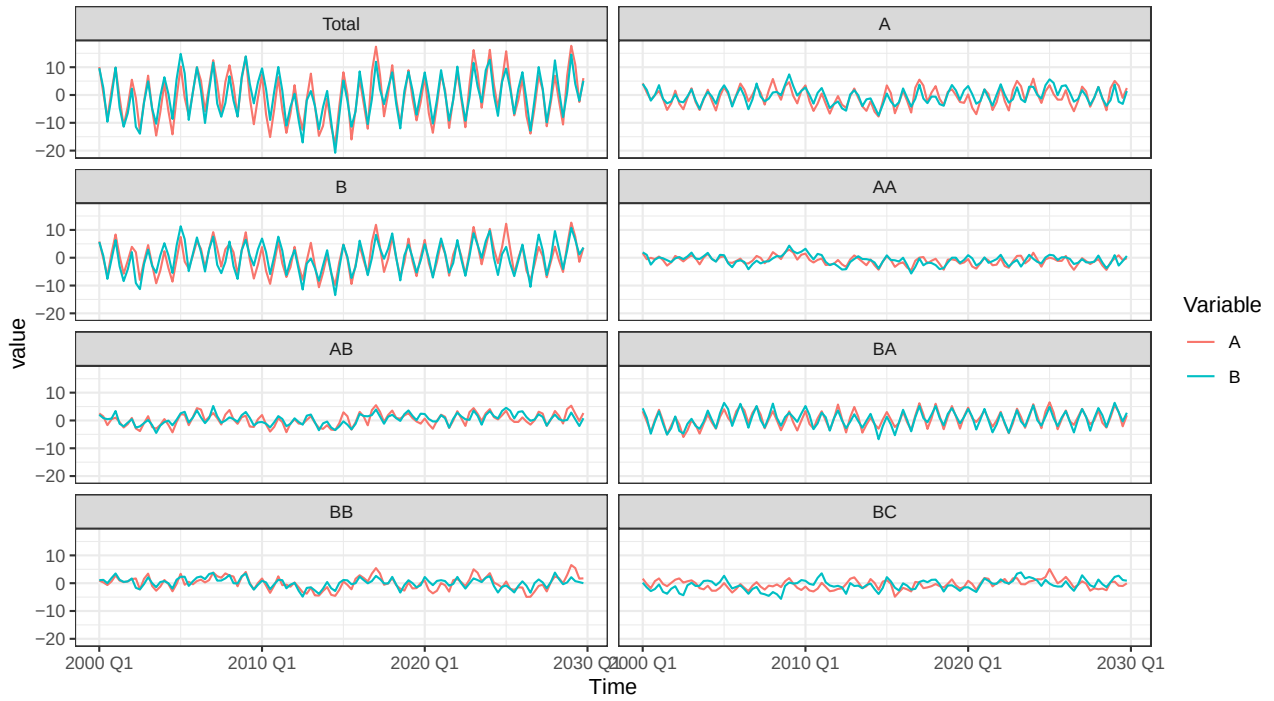


Figure S6: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 6.

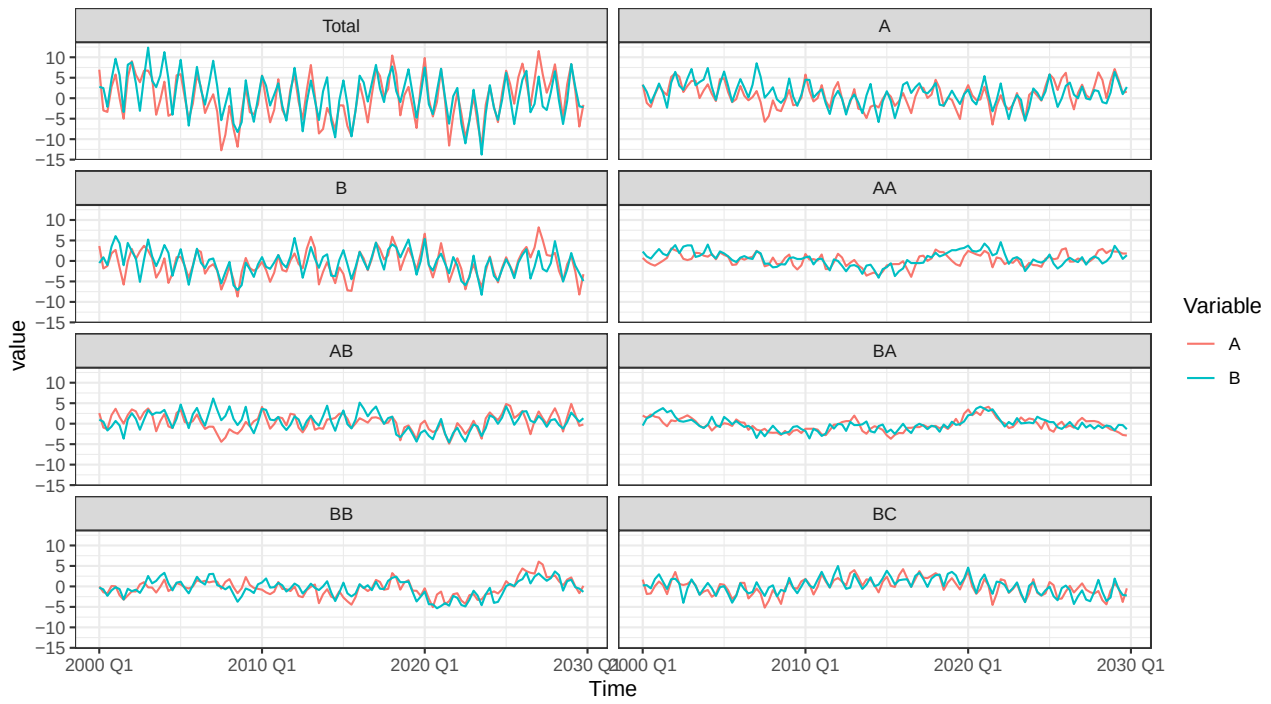


Figure S7: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 7.

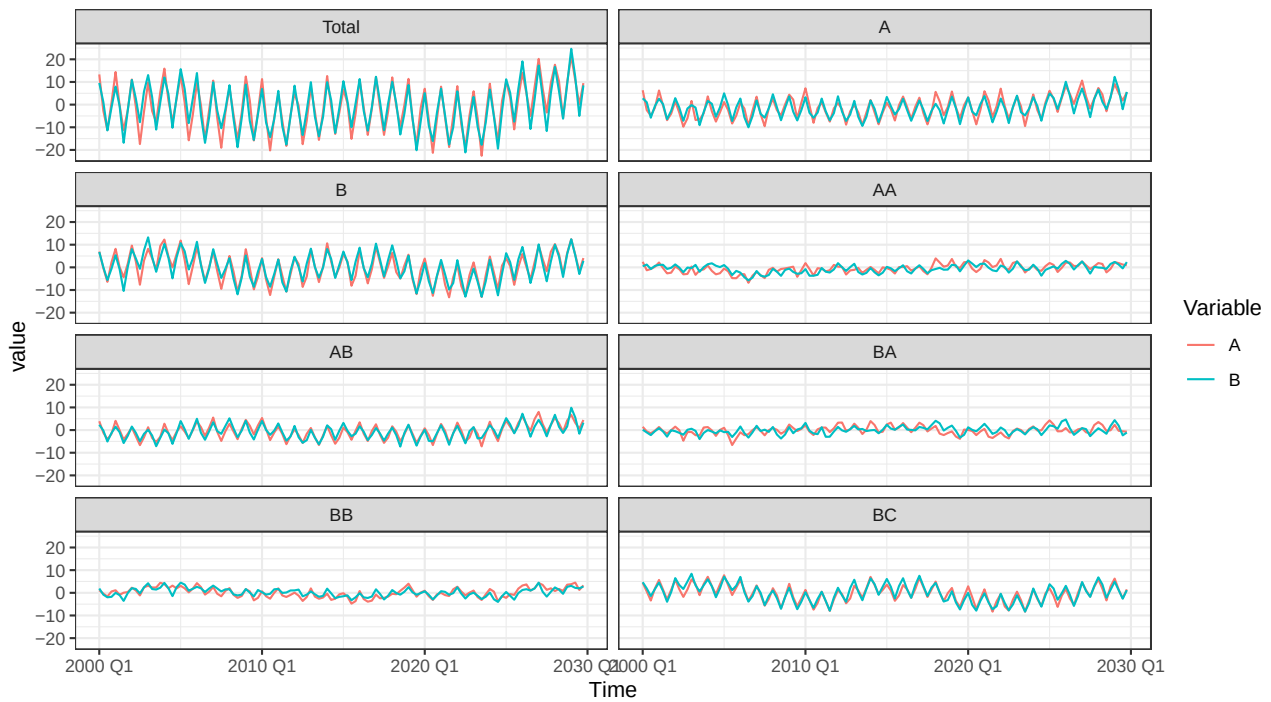


Figure S8: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 8.

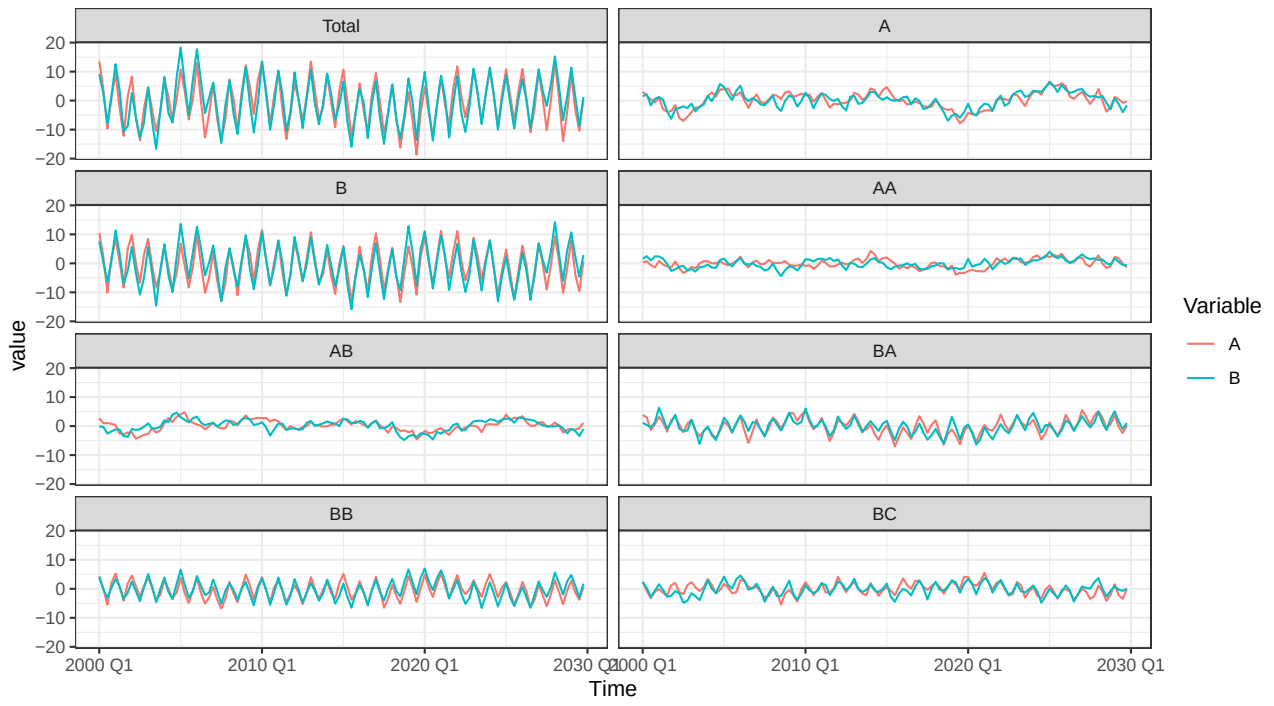


Figure S9: Simulation of a hierarchical multivariate time series, with the same structure as Figure 1, with $m = 2$, under Scenario 9.

Tables

Table S1: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 1. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.040	0.042	0.026	0.024	0.028	0.037	0.035	0.031	0.041	0.050	0.048	0.048
Total	2	0.029	0.036	0.018	0.018	0.024	0.038	0.033	0.025	0.026	0.040	0.036	0.039
A	1	0.028	0.036	0.030	0.035	0.037	0.031	0.033	0.035	0.040	0.040	0.033	0.038
A	2	0.033	0.038	0.033	0.025	0.035	0.034	0.031	0.041	0.041	0.043	0.041	0.043
B	1	0.029	0.032	0.024	0.021	0.034	0.030	0.033	0.041	0.040	0.036	0.035	0.045
B	2	0.024	0.031	0.026	0.025	0.036	0.041	0.032	0.039	0.036	0.043	0.044	0.036
AA	1	0.009	-0.003	0.009	0.009	0.006	0.010	0.016	0.018	0.015	0.014	0.018	0.009
AA	2	0.013	0.005	0.004	0.005	0.012	0.016	0.014	0.015	0.016	0.007	0.015	0.012
AB	1	0.013	0.009	0.020	0.008	0.016	0.013	0.014	0.006	0.014	0.012	0.020	0.013
AB	2	0.009	0.002	0.014	0.010	0.012	0.009	0.017	0.006	0.012	0.017	0.022	0.013
BA	1	0.012	0.005	0.014	0.011	0.012	0.002	0.002	0.003	0.007	0.009	0.013	0.006
BA	2	0.021	0.016	0.018	0.012	0.021	0.010	0.010	0.008	0.024	0.015	0.013	0.010
BB	1	0.009	0.008	0.017	0.009	0.018	0.007	0.014	0.001	0.011	0.012	0.021	0.011
BB	2	0.009	0.009	0.010	0.000	0.002	-0.002	0.007	-0.007	0.007	0.007	0.012	0.007
BC	1	0.004	0.011	0.010	0.004	0.012	0.008	0.011	0.001	0.008	0.006	0.008	0.006
BC	2	0.002	0.005	0.008	0.005	0.009	0.011	0.014	0.008	0.010	0.013	0.013	0.016
ETS													
Total	1	0.000	-0.000	0.000	0.000	0.002	-0.001	0.003	0.002	0.001	0.001	0.001	0.000
Total	2	0.005	0.005	0.004	0.003	0.005	0.003	0.005	0.004	0.002	0.004	0.003	0.002
A	1	0.003	0.001	-0.001	-0.001	0.002	0.001	0.001	0.000	0.000	0.001	0.001	-0.000
A	2	0.002	0.001	-0.001	-0.000	-0.000	-0.002	-0.002	-0.002	-0.002	-0.001	-0.003	-0.002
B	1	0.005	0.004	0.006	0.004	0.004	0.004	0.005	0.003	0.004	0.005	0.005	0.006
B	2	0.013	0.006	0.007	0.004	0.004	0.005	0.003	0.002	0.003	0.003	0.003	0.002
AA	1	0.005	0.001	0.002	0.001	-0.000	0.000	0.001	-0.000	0.001	0.001	0.001	0.001
AA	2	0.002	-0.001	0.003	0.002	-0.000	0.002	0.002	0.002	0.000	0.001	0.001	0.002
AB	1	0.004	0.002	0.004	0.002	0.000	0.000	-0.000	-0.000	-0.001	-0.001	0.001	0.001
AB	2	0.003	0.000	0.001	-0.000	-0.000	0.000	-0.002	-0.001	-0.000	-0.001	-0.001	-0.000
BA	1	0.003	0.002	0.001	-0.000	-0.001	0.001	-0.002	-0.001	0.000	-0.000	-0.001	-0.001
BA	2	0.000	0.001	-0.000	-0.002	-0.001	-0.000	-0.001	-0.001	0.000	-0.001	-0.002	-0.001
BB	1	0.005	0.001	-0.001	0.000	0.001	-0.002	-0.002	-0.001	-0.001	-0.002	-0.001	-0.001
BB	2	-0.001	0.002	-0.000	-0.001	-0.000	-0.000	-0.001	-0.001	0.000	0.000	-0.001	-0.001
BC	1	-0.000	0.002	0.001	0.000	0.001	0.001	0.001	0.001	0.001	-0.000	0.001	-0.001
BC	2	-0.000	-0.001	-0.001	-0.000	-0.001	-0.000	0.001	0.001	0.000	0.001	0.001	0.000
VAR													
Total	1	0.003	0.015	-0.008	0.001	0.005	0.013	-0.000	0.013	-0.014	0.008	-0.034	-0.006
Total	2	0.020	0.018	0.001	0.011	0.014	0.022	0.005	0.020	-0.002	0.007	-0.025	0.000
A	1	0.005	0.004	0.013	0.011	0.016	0.011	0.015	0.004	0.022	0.014	0.026	0.002
A	2	-0.002	0.005	0.005	0.002	-0.002	0.003	-0.001	0.008	0.001	-0.004	-0.011	-0.001
B	1	0.002	0.009	-0.005	-0.001	-0.006	0.003	-0.005	0.002	-0.015	0.001	-0.012	0.002
B	2	0.009	0.012	0.014	0.006	0.010	0.017	0.016	0.010	0.018	0.021	0.024	0.024
AA	1	0.001	0.002	0.002	-0.002	0.002	0.002	-0.001	-0.000	-0.002	0.004	0.003	0.003
AA	2	-0.021	-0.021	-0.007	-0.013	-0.023	-0.019	-0.007	-0.018	-0.012	-0.017	-0.004	-0.011
AB	1	-0.041	-0.046	-0.015	-0.023	-0.048	-0.051	-0.014	-0.050	-0.033	-0.044	0.005	-0.044
AB	2	-0.033	-0.031	-0.013	-0.024	-0.029	-0.032	-0.034	-0.046	-0.025	-0.030	-0.022	-0.037
BA	1	-0.054	-0.050	-0.032	-0.032	-0.056	-0.091	-0.063	-0.052	-0.034	-0.061	-0.047	-0.072
BA	2	0.004	-0.001	0.014	0.006	0.011	-0.006	-0.003	-0.007	0.016	-0.002	0.021	-0.002
BB	1	-0.005	-0.009	-0.023	-0.021	-0.033	-0.030	-0.027	-0.023	-0.027	-0.028	-0.026	-0.020
BB	2	-0.002	0.001	0.006	-0.008	0.008	-0.004	-0.004	-0.002	0.027	0.002	0.017	0.002
BC	1	-0.015	-0.003	-0.004	-0.005	0.001	-0.009	-0.005	-0.009	0.002	-0.000	-0.001	0.001
BC	2	0.014	-0.001	0.012	0.002	0.004	-0.016	-0.000	-0.011	0.010	-0.006	0.020	0.002

Table S2: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 2. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.039	0.032	0.030	0.048	0.038	0.046	0.056	0.047	0.041	0.048	0.050	0.045
Total	2	0.024	0.023	0.013	0.028	0.039	0.035	0.033	0.030	0.022	0.032	0.042	0.043
A	1	0.046	0.038	0.034	0.019	0.023	0.030	0.026	0.018	0.021	0.020	0.024	0.031
A	2	0.029	0.022	0.026	0.021	0.021	0.032	0.023	0.021	0.023	0.026	0.029	0.028
B	1	0.039	0.042	0.030	0.029	0.035	0.033	0.035	0.044	0.049	0.055	0.041	0.046
B	2	0.030	0.059	0.053	0.045	0.054	0.054	0.043	0.044	0.045	0.050	0.051	0.040
AA	1	0.036	0.030	0.023	0.027	0.035	0.028	0.034	0.035	0.046	0.032	0.042	0.037
AA	2	0.030	0.022	0.027	0.039	0.041	0.030	0.032	0.030	0.043	0.026	0.033	0.032
AB	1	0.022	0.024	0.027	0.017	0.027	0.023	0.031	0.017	0.031	0.020	0.025	0.015
AB	2	0.029	0.023	0.029	0.027	0.027	0.028	0.039	0.028	0.039	0.041	0.034	0.028
BA	1	0.022	0.018	0.023	0.016	0.027	0.019	0.022	0.014	0.021	0.015	0.027	0.018
BA	2	0.022	0.012	0.017	0.009	0.006	0.013	0.021	0.010	0.016	0.009	0.009	0.004
BB	1	0.020	0.018	0.019	0.013	0.010	0.019	0.016	0.016	0.022	0.020	0.023	0.013
BB	2	0.021	0.001	0.003	0.004	0.010	0.015	0.007	0.011	0.033	0.019	0.015	0.022
BC	1	0.029	0.032	0.026	0.024	0.038	0.030	0.016	0.026	0.033	0.021	0.021	0.021
BC	2	0.026	0.026	0.021	0.013	0.029	0.024	0.022	0.024	0.035	0.026	0.021	0.022
ETS													
Total	1	0.007	0.008	0.007	0.009	0.004	0.005	0.004	0.006	0.005	0.006	0.007	0.000
Total	2	0.013	0.010	0.007	0.009	0.008	0.010	0.006	0.008	0.012	0.012	0.013	0.002
A	1	0.010	0.004	0.003	0.001	0.001	0.004	0.005	0.003	0.005	0.009	0.004	0.006
A	2	0.007	0.009	0.003	-0.000	0.003	0.003	0.004	0.002	0.006	0.005	0.002	0.002
B	1	0.009	0.001	0.000	0.002	0.002	0.004	-0.000	0.001	0.004	0.003	-0.002	0.001
B	2	0.010	0.004	0.005	0.001	0.000	0.003	-0.000	0.001	0.003	0.004	-0.001	-0.001
AA	1	0.004	0.006	0.001	-0.000	0.001	0.002	-0.001	-0.001	0.001	-0.000	0.000	0.001
AA	2	0.003	0.003	0.003	-0.001	-0.000	0.001	-0.000	0.001	0.002	0.004	0.002	0.001
AB	1	0.000	-0.001	-0.003	-0.003	0.001	0.002	-0.002	-0.001	0.001	0.002	0.000	0.002
AB	2	-0.001	-0.004	-0.004	-0.004	-0.003	-0.002	-0.004	-0.002	-0.002	-0.001	-0.003	0.001
BA	1	0.003	0.005	0.001	-0.001	0.001	0.000	0.002	0.000	0.003	0.002	0.002	0.001
BA	2	0.005	0.006	0.002	0.001	0.001	-0.002	-0.000	-0.001	-0.001	-0.003	-0.001	0.003
BB	1	0.001	0.002	0.001	-0.000	0.003	0.000	0.000	0.001	0.001	-0.001	0.002	0.004
BB	2	0.001	0.004	0.003	0.002	0.005	0.004	0.003	0.002	0.000	0.000	0.005	0.005
BC	1	0.007	0.005	0.004	0.003	0.004	0.004	0.003	0.004	0.003	0.004	0.005	0.008
BC	2	0.006	0.003	0.002	0.003	0.003	0.000	0.001	0.002	0.002	0.003	0.003	0.007
VAR													
Total	1	0.003	0.016	-0.005	0.002	-0.016	0.020	0.015	0.003	-0.031	0.000	-0.026	-0.014
Total	2	0.011	0.026	0.005	0.014	0.013	0.025	0.020	0.012	-0.007	0.012	-0.027	-0.001
A	1	-0.008	-0.002	0.009	0.005	0.021	0.010	0.004	0.004	0.026	0.006	0.020	0.009
A	2	0.003	0.011	-0.000	-0.001	0.002	0.003	-0.002	-0.007	-0.007	-0.003	-0.012	-0.006
B	1	0.001	0.013	0.001	0.004	0.003	0.012	0.004	0.005	-0.009	-0.004	-0.021	-0.007
B	2	0.018	0.011	0.009	0.010	0.021	0.021	0.017	0.021	0.026	0.027	0.035	0.024
AA	1	-0.004	0.002	0.000	-0.002	0.001	-0.001	-0.004	-0.005	-0.000	-0.007	-0.003	-0.004
AA	2	-0.006	-0.013	-0.008	-0.012	-0.011	-0.024	-0.005	-0.011	-0.009	-0.022	-0.003	-0.022
AB	1	-0.083	-0.099	-0.049	-0.038	-0.060	-0.100	-0.078	-0.061	-0.028	-0.073	-0.025	-0.024
AB	2	-0.033	-0.046	-0.016	-0.023	-0.028	-0.046	-0.031	-0.026	-0.038	-0.021	0.010	-0.024
BA	1	-0.076	-0.096	-0.053	-0.062	-0.106	-0.131	-0.107	-0.114	-0.095	-0.110	-0.029	-0.070
BA	2	-0.009	-0.009	0.005	-0.016	-0.009	-0.019	0.003	-0.010	0.004	-0.010	0.030	-0.002
BB	1	-0.017	-0.015	-0.011	-0.016	-0.025	-0.021	-0.012	-0.013	-0.018	-0.024	-0.018	-0.022
BB	2	0.011	-0.018	0.004	-0.021	-0.012	-0.039	-0.014	-0.025	-0.011	-0.013	0.029	0.003
BC	1	0.007	0.005	-0.003	-0.007	-0.005	-0.003	-0.004	-0.011	-0.013	-0.008	0.003	-0.007
BC	2	-0.002	-0.009	0.001	-0.006	-0.002	-0.020	-0.002	-0.014	0.004	-0.010	0.024	0.001

Table S3: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 3. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.036	0.034	0.048	0.052	0.071	0.066	0.062	0.074	0.079	0.076	0.070	0.079
Total	2	0.035	0.037	0.041	0.041	0.057	0.059	0.061	0.061	0.067	0.070	0.069	0.069
A	1	0.043	0.039	0.057	0.042	0.057	0.054	0.061	0.048	0.060	0.063	0.064	0.048
A	2	0.036	0.051	0.042	0.041	0.048	0.058	0.059	0.053	0.065	0.076	0.067	0.064
B	1	0.061	0.059	0.049	0.043	0.055	0.064	0.058	0.060	0.069	0.070	0.063	0.064
B	2	0.046	0.060	0.048	0.045	0.043	0.055	0.050	0.051	0.047	0.053	0.045	0.047
AA	1	0.014	0.013	0.010	0.009	0.015	0.012	0.010	0.013	0.018	0.011	0.006	0.010
AA	2	0.009	0.003	0.008	0.004	0.009	0.000	0.006	0.003	0.005	-0.006	-0.002	-0.001
AB	1	0.013	0.015	0.014	0.009	0.009	0.013	0.017	0.009	0.012	0.013	0.014	0.014
AB	2	0.019	0.010	0.008	0.007	0.007	0.007	0.005	0.018	0.013	0.003	0.008	0.018
BA	1	0.000	0.008	0.009	0.004	0.003	0.006	0.014	0.010	0.011	0.008	0.009	0.006
BA	2	0.006	0.008	0.009	0.003	0.004	0.007	0.010	0.013	0.009	0.012	0.014	0.006
BB	1	0.005	0.002	0.001	-0.006	-0.004	0.004	0.004	0.001	0.007	0.009	0.011	0.008
BB	2	0.013	0.015	0.013	0.007	0.015	0.014	0.007	0.005	0.015	0.011	0.010	0.008
BC	1	0.004	-0.001	0.003	0.003	0.002	-0.006	-0.003	-0.003	0.004	-0.001	0.002	0.001
BC	2	0.007	0.006	0.009	0.010	0.005	0.003	0.005	0.002	0.009	0.005	0.008	0.008
ETS													
Total	1	0.003	0.006	0.007	0.007	0.005	0.007	0.006	0.004	0.002	0.002	0.002	0.003
Total	2	0.007	0.007	0.005	0.005	0.003	0.004	0.005	0.004	0.004	0.002	0.003	0.003
A	1	0.007	0.005	0.002	0.003	0.003	0.003	0.002	0.001	0.001	0.001	0.001	0.002
A	2	0.004	0.006	0.003	0.004	0.003	0.004	0.003	0.003	0.002	0.003	0.003	0.004
B	1	0.004	0.004	0.002	0.002	0.002	0.003	0.000	0.001	0.001	0.001	0.000	0.001
B	2	0.001	0.004	0.002	0.001	0.001	0.001	-0.001	0.000	0.000	0.001	0.000	-0.001
AA	1	0.001	-0.002	0.001	-0.001	-0.000	-0.002	-0.001	-0.001	-0.000	-0.001	-0.001	-0.001
AA	2	0.005	-0.001	0.001	-0.000	0.001	-0.001	-0.000	-0.001	0.001	-0.000	0.001	-0.000
AB	1	0.005	0.004	0.002	0.002	0.002	0.001	0.000	0.001	0.002	0.001	-0.000	-0.000
AB	2	0.003	0.002	0.002	0.001	0.002	0.000	-0.000	0.000	0.000	-0.000	-0.000	-0.001
BA	1	0.002	0.001	0.002	0.002	0.002	-0.000	0.002	0.002	0.002	0.002	0.001	0.001
BA	2	0.001	-0.004	-0.001	-0.001	-0.001	-0.002	-0.001	-0.001	-0.001	-0.000	-0.000	-0.000
BB	1	0.001	-0.001	-0.000	-0.001	0.000	-0.001	-0.001	-0.001	-0.000	-0.000	-0.001	-0.001
BB	2	0.003	-0.001	0.000	0.000	0.000	-0.000	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
BC	1	-0.002	-0.003	-0.003	-0.003	-0.002	-0.003	-0.002	-0.002	-0.001	-0.002	-0.001	-0.002
BC	2	0.001	-0.001	0.000	0.000	0.000	0.000	0.000	-0.000	0.000	-0.001	-0.000	-0.000
VAR													
Total	1	-0.008	-0.004	-0.015	0.001	0.001	0.004	-0.015	0.004	0.014	0.030	-0.001	0.017
Total	2	0.013	0.013	0.002	0.012	0.004	0.023	0.012	0.017	0.015	0.035	-0.001	0.022
A	1	0.008	0.009	0.017	0.010	0.006	-0.003	0.009	0.001	0.016	-0.002	0.015	-0.005
A	2	0.011	-0.000	-0.004	-0.002	0.003	0.000	0.001	0.002	0.003	0.004	-0.002	0.005
B	1	0.010	0.010	-0.001	0.000	-0.002	0.003	-0.005	-0.002	-0.006	0.004	-0.011	-0.003
B	2	0.016	0.025	0.017	0.012	0.018	0.021	0.022	0.015	0.010	0.009	0.015	0.008
AA	1	-0.001	0.003	0.001	-0.003	-0.003	-0.002	-0.001	-0.002	-0.004	-0.000	-0.002	0.000
AA	2	-0.004	-0.002	-0.002	-0.008	-0.002	-0.013	0.003	-0.005	0.005	-0.006	0.005	0.001
AB	1	-0.032	-0.052	-0.029	-0.021	-0.039	-0.057	-0.017	-0.026	-0.023	-0.046	-0.018	-0.035
AB	2	-0.018	-0.014	-0.004	-0.020	-0.022	-0.037	-0.019	-0.007	-0.011	-0.033	-0.008	-0.015
BA	1	-0.040	-0.088	-0.046	-0.041	-0.054	-0.085	-0.061	-0.055	-0.032	-0.058	-0.028	-0.047
BA	2	-0.012	-0.008	0.015	-0.000	0.014	-0.010	0.010	-0.001	0.013	-0.009	0.014	-0.011
BB	1	-0.015	-0.019	-0.018	-0.022	-0.019	-0.023	-0.020	-0.019	-0.015	-0.023	-0.019	-0.020
BB	2	-0.001	-0.008	0.003	-0.003	0.004	-0.016	0.007	-0.006	0.014	-0.016	0.013	-0.004
BC	1	-0.006	-0.009	-0.005	-0.011	-0.006	-0.012	-0.015	-0.010	0.011	-0.000	0.002	-0.007
BC	2	-0.003	-0.012	0.005	-0.005	0.001	-0.019	0.004	-0.004	0.021	-0.008	0.020	-0.007

Table S4: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 4. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.066	0.054	0.059	0.043	0.064	0.071	0.069	0.066	0.069	0.067	0.057	0.065
Total	2	0.066	0.056	0.069	0.050	0.053	0.053	0.043	0.038	0.035	0.047	0.042	0.047
A	1	0.048	0.059	0.032	0.028	0.032	0.040	0.038	0.028	0.038	0.029	0.037	0.048
A	2	0.034	0.046	0.034	0.030	0.029	0.026	0.027	0.024	0.025	0.030	0.029	0.040
B	1	0.032	0.041	0.041	0.042	0.045	0.053	0.038	0.060	0.053	0.048	0.053	0.061
B	2	0.036	0.048	0.061	0.056	0.066	0.064	0.060	0.076	0.063	0.072	0.080	0.071
AA	1	0.036	0.027	0.026	0.020	0.041	0.032	0.029	0.033	0.039	0.036	0.039	0.029
AA	2	0.038	0.026	0.035	0.031	0.040	0.033	0.033	0.036	0.043	0.028	0.035	0.032
AB	1	0.027	0.027	0.020	0.020	0.042	0.028	0.020	0.025	0.036	0.028	0.026	0.024
AB	2	0.029	0.024	0.030	0.034	0.045	0.034	0.032	0.031	0.048	0.036	0.030	0.023
BA	1	0.041	0.035	0.033	0.022	0.026	0.020	0.026	0.021	0.021	0.019	0.024	0.018
BA	2	0.034	0.028	0.030	0.033	0.035	0.032	0.038	0.029	0.035	0.030	0.034	0.031
BB	1	0.041	0.022	0.028	0.020	0.023	0.017	0.011	0.012	0.024	0.014	0.023	0.015
BB	2	0.022	0.019	0.025	0.007	0.008	0.012	0.008	0.002	0.010	0.011	0.023	0.012
BC	1	0.027	0.028	0.025	0.025	0.031	0.019	0.023	0.015	0.023	0.018	0.018	0.017
BC	2	0.041	0.028	0.022	0.016	0.019	0.014	0.018	0.009	0.020	0.021	0.030	0.021
ETS													
Total	1	0.018	0.010	0.006	0.004	0.003	0.008	0.006	0.003	0.001	0.001	0.002	0.003
Total	2	0.017	0.012	0.007	0.002	0.002	0.006	0.004	0.002	0.002	0.001	0.001	0.003
A	1	0.018	0.015	0.007	0.006	0.004	0.004	0.004	0.003	0.002	0.005	0.003	0.004
A	2	0.017	0.020	0.012	0.008	0.005	0.006	0.006	0.006	0.006	0.008	0.006	0.006
B	1	0.014	0.007	0.003	0.002	-0.000	-0.001	-0.003	-0.001	-0.001	-0.001	-0.001	-0.004
B	2	0.010	0.007	0.005	0.002	0.000	-0.000	-0.001	-0.000	0.002	-0.000	0.001	-0.002
AA	1	0.006	0.008	0.006	0.004	0.004	0.002	0.003	0.003	0.003	0.003	0.002	0.001
AA	2	0.013	0.006	0.003	0.002	0.005	0.001	0.001	0.002	0.002	0.001	0.001	0.001
AB	1	0.003	0.002	0.001	-0.002	0.000	-0.000	-0.001	-0.001	-0.001	-0.002	0.001	-0.001
AB	2	0.004	0.000	0.003	-0.000	0.000	-0.001	0.001	-0.001	-0.000	0.001	0.002	-0.000
BA	1	0.007	0.005	0.005	0.003	0.003	0.002	0.001	0.001	0.001	0.001	0.002	0.002
BA	2	0.011	0.004	0.004	0.003	0.003	0.003	0.003	0.002	0.001	0.004	0.003	0.003
BB	1	0.007	0.001	0.002	-0.000	0.000	0.001	-0.000	-0.000	0.001	0.002	0.001	0.001
BB	2	0.014	0.006	0.006	0.004	0.002	0.003	0.002	0.001	0.002	0.003	0.001	0.001
BC	1	0.011	0.010	0.008	0.007	0.006	0.004	0.007	0.003	0.005	0.005	0.005	0.003
BC	2	0.012	0.009	0.005	0.004	0.003	0.004	0.004	0.003	0.003	0.004	0.003	0.002
VAR													
Total	1	0.007	0.001	-0.011	-0.013	0.001	0.018	0.005	0.012	-0.001	0.024	0.002	0.018
Total	2	0.025	0.021	0.011	0.019	0.020	0.036	0.016	0.023	0.010	0.043	0.011	0.027
A	1	0.006	0.008	0.016	0.011	0.017	0.010	0.011	0.009	0.019	0.010	0.018	0.004
A	2	0.005	0.012	-0.003	-0.000	-0.002	0.000	-0.004	-0.005	-0.003	0.011	0.004	0.006
B	1	0.008	0.007	-0.001	0.009	0.006	0.012	0.004	0.006	-0.004	0.003	-0.007	-0.001
B	2	0.017	0.027	0.022	0.025	0.025	0.024	0.024	0.016	0.018	0.012	0.017	0.014
AA	1	0.008	0.003	0.006	0.002	0.002	-0.001	-0.007	-0.003	-0.008	-0.013	-0.011	-0.010
AA	2	-0.009	-0.024	-0.012	-0.017	-0.019	-0.021	-0.009	-0.017	-0.005	-0.003	0.004	-0.009
AB	1	-0.070	-0.065	-0.045	-0.051	-0.040	-0.064	-0.071	-0.044	-0.015	-0.063	-0.066	-0.069
AB	2	-0.023	-0.046	-0.025	-0.006	0.005	-0.012	-0.014	-0.022	0.013	-0.037	-0.024	-0.042
BA	1	-0.073	-0.081	-0.059	-0.071	-0.094	-0.125	-0.100	-0.078	-0.071	-0.073	-0.039	-0.086
BA	2	-0.010	-0.016	-0.009	-0.021	-0.020	-0.014	0.009	-0.012	0.004	-0.008	0.022	-0.010
BB	1	-0.018	-0.024	-0.019	-0.020	-0.022	-0.024	-0.019	-0.017	-0.026	-0.029	-0.015	-0.022
BB	2	-0.011	-0.018	-0.001	-0.018	-0.010	-0.014	0.002	-0.025	-0.001	-0.015	0.008	-0.006
BC	1	-0.001	-0.007	-0.012	-0.001	0.006	-0.003	-0.003	-0.006	-0.001	0.003	0.001	-0.004
BC	2	-0.000	-0.001	0.002	-0.012	-0.001	-0.011	0.017	0.005	0.016	-0.000	0.028	0.000

Table S5: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 5. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.083	0.090	0.094	0.079	0.061	0.052	0.044	0.054	0.061	0.056	0.060	0.061
Total	2	0.091	0.092	0.091	0.077	0.048	0.055	0.058	0.055	0.063	0.064	0.066	0.060
A	1	0.052	0.057	0.061	0.068	0.059	0.052	0.045	0.036	0.041	0.039	0.035	0.037
A	2	0.067	0.058	0.053	0.049	0.040	0.031	0.035	0.027	0.039	0.038	0.037	0.030
B	1	0.063	0.045	0.048	0.035	0.035	0.046	0.051	0.048	0.041	0.046	0.049	0.040
B	2	0.061	0.047	0.057	0.051	0.053	0.054	0.048	0.039	0.033	0.043	0.040	0.025
AA	1	0.053	0.043	0.037	0.017	0.040	0.030	0.038	0.031	0.036	0.025	0.035	0.020
AA	2	0.042	0.031	0.040	0.028	0.042	0.023	0.026	0.013	0.030	0.012	0.026	0.016
AB	1	0.036	0.043	0.039	0.022	0.036	0.027	0.046	0.022	0.026	0.006	0.028	0.015
AB	2	0.051	0.049	0.032	0.038	0.047	0.045	0.058	0.043	0.033	0.030	0.044	0.022
BA	1	0.026	0.024	0.028	0.014	0.015	0.016	0.036	0.006	0.006	0.007	0.017	0.010
BA	2	0.033	0.026	0.023	0.022	0.033	0.016	0.021	0.012	0.015	0.010	0.020	0.011
BB	1	0.026	0.020	0.025	0.022	0.029	0.016	0.027	0.018	0.019	0.024	0.032	0.017
BB	2	0.053	0.023	0.029	0.026	0.039	0.040	0.033	0.042	0.048	0.032	0.047	0.020
BC	1	0.043	0.020	0.037	0.025	0.021	0.021	0.021	0.012	0.011	0.002	0.008	0.000
BC	2	0.042	0.019	0.019	0.013	0.031	0.016	0.021	0.011	0.007	0.001	0.005	-0.003
ETS													
Total	1	0.023	0.019	0.008	-0.004	0.001	0.002	-0.003	-0.005	0.001	0.004	-0.000	-0.006
Total	2	0.010	0.014	0.011	-0.001	0.004	0.002	0.002	0.000	0.011	0.009	0.008	-0.001
A	1	0.019	0.011	0.006	0.006	0.009	0.007	0.006	0.000	0.002	0.001	0.002	0.002
A	2	0.030	0.020	0.016	0.011	0.009	0.010	0.008	0.002	0.006	0.008	0.006	0.001
B	1	0.007	0.003	0.005	0.000	0.005	-0.000	0.003	-0.000	0.004	0.006	0.006	0.002
B	2	0.006	0.005	0.005	-0.001	0.010	0.007	0.005	0.003	0.009	0.010	0.011	0.006
AA	1	0.013	0.005	0.006	0.005	0.007	0.008	0.007	0.005	0.004	0.009	0.006	0.005
AA	2	0.014	0.013	0.007	0.003	0.006	0.006	0.004	0.006	0.008	0.006	0.009	0.012
AB	1	0.020	0.011	0.011	0.007	0.006	0.006	0.007	0.004	0.009	0.006	0.007	0.003
AB	2	0.019	0.009	0.009	0.005	0.007	0.008	0.003	0.002	0.011	0.004	0.001	0.004
BA	1	0.019	0.014	0.007	0.007	0.004	0.005	0.005	0.002	0.003	0.002	0.003	0.005
BA	2	0.019	0.014	0.006	0.010	0.002	0.005	0.002	0.003	0.002	0.003	0.003	0.006
BB	1	0.012	0.009	0.006	0.009	0.007	0.005	0.005	0.004	0.001	-0.003	0.001	0.002
BB	2	0.018	0.007	0.001	0.005	-0.000	0.004	0.002	0.003	-0.001	-0.001	-0.002	0.001
BC	1	0.022	0.013	0.008	0.010	0.007	0.005	0.004	0.004	0.002	0.000	-0.003	0.003
BC	2	0.023	0.013	0.011	0.008	0.008	0.004	0.007	0.004	0.001	0.001	0.001	0.005
VAR													
Total	1	0.026	0.051	0.021	0.012	0.007	0.018	-0.006	-0.006	-0.021	0.021	-0.015	-0.001
Total	2	0.034	0.052	0.043	0.031	0.037	0.060	0.046	0.026	0.021	0.047	-0.002	0.025
A	1	-0.024	-0.010	0.004	-0.010	-0.017	-0.004	0.015	0.003	0.012	0.003	0.011	-0.003
A	2	0.028	0.017	0.010	0.004	0.001	0.008	0.002	0.000	-0.008	-0.007	-0.008	-0.008
B	1	0.012	0.008	0.012	0.010	0.008	0.009	0.000	0.003	-0.001	0.016	0.001	0.009
B	2	0.020	0.018	0.021	0.033	0.038	0.040	0.026	0.041	0.041	0.035	0.042	0.027
AA	1	-0.006	0.001	0.006	0.004	-0.001	0.003	0.007	0.002	-0.009	-0.003	-0.002	-0.001
AA	2	-0.017	-0.020	-0.018	-0.025	-0.028	-0.034	-0.017	-0.027	-0.019	-0.034	-0.025	-0.034
AB	1	-0.073	-0.082	-0.066	-0.087	-0.080	-0.128	-0.074	-0.061	-0.067	-0.088	-0.064	-0.062
AB	2	-0.023	-0.058	-0.035	-0.016	-0.008	-0.044	-0.028	-0.033	-0.014	-0.052	-0.030	-0.028
BA	1	-0.077	-0.110	-0.104	-0.108	-0.129	-0.133	-0.111	-0.115	-0.146	-0.204	-0.148	-0.141
BA	2	0.003	-0.017	-0.010	-0.019	-0.003	-0.010	-0.005	-0.006	0.007	-0.017	0.019	-0.012
BB	1	-0.028	-0.021	-0.027	-0.027	-0.021	-0.032	-0.047	-0.028	-0.044	-0.043	-0.024	-0.024
BB	2	0.019	-0.014	-0.009	-0.006	0.002	-0.024	-0.027	-0.005	0.016	-0.027	0.005	-0.004
BC	1	-0.005	0.000	0.005	0.000	-0.003	-0.009	-0.005	-0.008	0.008	0.007	0.017	0.001
BC	2	0.006	0.005	0.016	-0.019	-0.005	-0.009	0.000	-0.028	-0.002	-0.028	0.023	-0.005

Table S6: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 6. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.096	0.105	0.082	0.086	0.100	0.120	0.108	0.099	0.116	0.123	0.109	0.112
Total	2	0.092	0.108	0.096	0.075	0.091	0.104	0.105	0.101	0.110	0.106	0.106	0.109
A	1	0.074	0.070	0.070	0.078	0.088	0.092	0.094	0.096	0.096	0.098	0.097	0.098
A	2	0.076	0.077	0.070	0.086	0.102	0.098	0.088	0.084	0.095	0.097	0.094	0.097
B	1	0.089	0.100	0.094	0.080	0.096	0.112	0.096	0.088	0.095	0.106	0.092	0.089
B	2	0.099	0.084	0.107	0.097	0.114	0.100	0.107	0.098	0.102	0.100	0.103	0.098
AA	1	0.046	0.048	0.040	0.029	0.038	0.042	0.039	0.028	0.034	0.036	0.031	0.019
AA	2	0.038	0.043	0.034	0.019	0.036	0.032	0.031	0.023	0.042	0.030	0.029	0.026
AB	1	0.046	0.046	0.036	0.036	0.044	0.027	0.035	0.025	0.033	0.022	0.032	0.036
AB	2	0.048	0.042	0.044	0.048	0.056	0.046	0.050	0.058	0.064	0.049	0.044	0.048
BA	1	0.052	0.040	0.041	0.030	0.033	0.031	0.034	0.029	0.037	0.035	0.030	0.028
BA	2	0.036	0.029	0.035	0.023	0.034	0.029	0.034	0.033	0.031	0.028	0.039	0.037
BB	1	0.041	0.033	0.039	0.031	0.047	0.024	0.026	0.025	0.036	0.020	0.025	0.022
BB	2	0.036	0.037	0.039	0.035	0.035	0.033	0.035	0.030	0.036	0.029	0.030	0.024
BC	1	0.031	0.026	0.026	0.021	0.034	0.027	0.020	0.018	0.023	0.027	0.027	0.027
BC	2	0.038	0.030	0.040	0.023	0.034	0.033	0.023	0.020	0.028	0.033	0.023	0.025
ETS													
Total	1	0.010	0.012	0.017	0.009	0.009	0.010	0.009	0.006	0.006	0.004	0.004	0.003
Total	2	0.013	0.011	0.015	0.009	0.008	0.008	0.006	0.005	0.003	0.001	0.002	0.002
A	1	0.013	0.013	0.008	0.007	0.007	0.006	0.005	0.006	0.007	0.006	0.006	0.005
A	2	0.015	0.017	0.014	0.009	0.013	0.014	0.009	0.008	0.006	0.006	0.005	0.003
B	1	0.019	0.010	0.007	0.004	0.006	0.004	0.003	0.002	0.003	0.001	0.002	0.001
B	2	0.012	0.003	0.004	0.005	0.005	0.001	0.002	0.003	0.001	-0.001	0.000	0.002
AA	1	0.018	0.013	0.008	0.007	0.008	0.007	0.005	0.006	0.005	0.007	0.005	0.003
AA	2	0.011	0.007	0.004	0.003	0.002	0.002	0.001	0.001	0.002	0.002	0.001	0.002
AB	1	0.014	0.007	0.005	0.004	0.006	0.004	0.002	0.003	0.005	0.004	0.003	0.003
AB	2	0.013	0.008	0.005	0.005	0.008	0.005	0.005	0.004	0.005	0.005	0.004	0.004
BA	1	0.010	0.011	0.009	0.007	0.006	0.006	0.006	0.004	0.004	0.005	0.005	0.004
BA	2	0.012	0.005	0.004	0.006	0.006	0.003	0.003	0.002	0.004	0.003	0.002	0.002
BB	1	0.016	0.004	0.000	0.005	0.005	0.000	0.003	0.004	0.004	0.002	0.003	0.003
BB	2	0.016	0.007	0.005	0.003	0.004	0.003	0.005	0.004	0.004	0.002	0.003	0.002
BC	1	0.014	0.004	0.001	-0.000	0.001	0.000	0.001	0.002	0.003	0.002	0.002	0.000
BC	2	0.015	0.008	0.006	0.002	0.005	0.001	0.000	0.003	0.005	0.003	0.003	0.001
VAR													
Total	1	-0.004	-0.012	-0.006	0.009	-0.004	-0.012	-0.025	0.000	-0.006	0.016	0.007	0.003
Total	2	0.022	0.021	0.023	0.019	0.009	0.016	-0.009	0.011	0.009	0.026	0.010	0.026
A	1	0.007	0.000	0.008	-0.010	0.007	-0.004	-0.005	-0.002	0.008	0.000	0.012	-0.001
A	2	0.009	0.010	0.006	0.004	-0.003	-0.005	-0.012	-0.003	0.000	0.002	-0.003	0.007
B	1	0.006	0.010	0.008	0.009	0.000	0.004	0.001	0.004	-0.002	0.011	0.000	0.002
B	2	0.029	0.022	0.017	0.024	0.028	0.024	0.026	0.019	0.017	0.012	0.018	0.018
AA	1	0.004	-0.002	-0.009	-0.009	-0.006	-0.007	-0.012	-0.015	-0.013	-0.009	-0.010	-0.010
AA	2	-0.012	-0.017	-0.015	-0.024	-0.022	-0.015	-0.010	-0.016	-0.003	-0.013	-0.002	-0.009
AB	1	-0.049	-0.075	-0.067	-0.022	-0.017	-0.050	-0.047	-0.050	-0.031	-0.045	-0.030	-0.034
AB	2	-0.043	-0.059	-0.005	-0.014	-0.037	-0.031	-0.008	-0.012	-0.027	-0.017	-0.008	-0.038
BA	1	-0.067	-0.094	-0.066	-0.070	-0.079	-0.125	-0.071	-0.068	-0.039	-0.068	-0.056	-0.064
BA	2	-0.016	-0.022	-0.011	-0.022	-0.006	-0.014	0.011	-0.002	0.006	-0.012	0.007	-0.008
BB	1	-0.018	-0.023	-0.020	-0.019	-0.023	-0.022	-0.021	-0.026	-0.010	-0.004	-0.014	-0.018
BB	2	-0.004	-0.030	-0.018	-0.019	0.002	-0.015	-0.002	-0.007	0.014	-0.010	0.006	-0.010
BC	1	0.000	-0.007	-0.000	-0.004	0.002	-0.003	0.003	-0.004	0.000	-0.006	-0.004	-0.008
BC	2	-0.005	-0.015	-0.005	-0.004	0.002	-0.006	0.023	0.006	0.015	-0.004	0.013	-0.003

Table S7: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 7. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.037	0.038	0.022	0.021	0.030	0.039	0.045	0.045	0.041	0.045	0.034	0.046
Total	2	0.046	0.049	0.036	0.041	0.047	0.042	0.037	0.046	0.049	0.043	0.040	0.045
A	1	0.037	0.044	0.028	0.029	0.037	0.043	0.041	0.038	0.037	0.041	0.038	0.045
A	2	0.029	0.033	0.032	0.031	0.032	0.030	0.037	0.035	0.043	0.031	0.037	0.038
B	1	0.047	0.055	0.040	0.023	0.025	0.021	0.030	0.032	0.031	0.032	0.036	0.035
B	2	0.045	0.061	0.051	0.042	0.043	0.047	0.044	0.039	0.041	0.044	0.049	0.048
AA	1	0.019	0.015	0.016	0.007	0.007	0.011	0.012	0.009	0.011	0.015	0.014	0.008
AA	2	0.013	0.014	0.022	0.010	0.008	0.008	0.012	0.006	0.001	0.006	0.010	0.006
AB	1	0.009	0.000	0.012	0.005	0.013	0.004	0.010	0.002	0.011	0.005	0.011	0.004
AB	2	0.014	0.007	0.009	0.008	0.013	0.009	0.010	0.002	0.005	0.011	0.007	-0.000
BA	1	0.014	-0.001	0.004	0.005	0.003	0.006	0.006	0.007	0.007	0.004	0.004	0.006
BA	2	0.017	0.010	0.006	0.005	0.009	0.004	0.009	0.010	0.011	0.011	0.006	0.008
BB	1	0.018	0.014	0.013	0.017	0.021	0.007	0.017	0.013	0.014	0.012	0.015	0.005
BB	2	0.008	0.005	0.004	0.007	0.013	0.009	0.016	0.013	0.012	0.010	0.017	0.010
BC	1	0.006	0.017	0.020	0.017	0.014	0.014	0.014	0.003	0.010	0.009	0.010	0.002
BC	2	0.007	0.010	0.005	0.010	0.009	0.008	0.006	0.009	0.008	0.005	0.006	0.003
ETS													
Total	1	0.001	0.004	0.004	0.003	0.004	0.004	0.002	0.001	0.001	0.001	0.001	-0.000
Total	2	0.001	0.004	0.003	0.002	0.004	0.003	0.001	0.002	0.001	0.001	0.000	-0.001
A	1	0.011	0.005	0.003	0.001	0.001	0.001	0.000	0.000	0.001	0.000	-0.000	0.000
A	2	0.009	0.003	0.002	0.001	0.001	0.002	0.002	0.002	0.001	0.001	0.001	0.001
B	1	0.006	0.005	0.003	0.001	0.001	0.002	0.000	-0.000	0.002	0.001	0.002	0.000
B	2	0.013	0.007	0.005	0.004	0.005	0.004	0.003	0.002	0.003	0.002	0.001	0.000
AA	1	0.008	0.005	0.005	0.004	0.004	0.001	0.002	0.002	0.003	0.002	0.003	0.002
AA	2	0.005	0.004	0.003	0.003	0.002	0.001	0.001	-0.000	0.001	0.000	0.002	0.001
AB	1	0.004	0.003	0.002	0.003	0.003	0.002	0.002	0.002	0.002	0.002	0.002	0.000
AB	2	0.005	0.003	0.003	0.003	0.003	0.002	0.003	0.003	0.004	0.004	0.003	0.002
BA	1	0.004	0.000	-0.002	-0.000	-0.000	0.000	-0.001	-0.000	0.000	0.001	-0.001	0.000
BA	2	0.002	0.002	0.001	0.001	0.000	0.000	0.002	0.001	-0.000	0.002	0.001	0.001
BB	1	0.005	0.004	0.001	0.001	-0.000	-0.000	-0.000	0.001	-0.000	0.001	0.000	0.000
BB	2	0.004	0.003	0.001	0.002	-0.000	0.001	0.001	0.002	0.001	0.003	0.001	0.002
BC	1	0.003	0.004	0.004	-0.000	0.000	0.000	0.002	0.000	0.001	0.001	0.000	-0.000
BC	2	0.002	0.004	0.002	0.002	0.000	-0.001	-0.001	-0.000	0.000	0.000	0.000	-0.000
VAR													
Total	1	-0.013	-0.007	0.009	-0.002	-0.005	0.005	0.003	0.001	-0.026	0.004	-0.006	0.005
Total	2	0.014	0.019	0.019	0.016	0.013	0.022	0.007	0.009	-0.014	0.011	-0.003	0.017
A	1	0.016	0.010	0.011	0.016	0.026	0.023	0.026	0.018	0.025	0.020	0.021	0.009
A	2	0.009	0.001	0.005	-0.002	-0.002	0.002	0.001	-0.002	-0.007	-0.004	-0.003	0.002
B	1	-0.000	0.012	-0.002	-0.005	-0.005	0.004	-0.007	-0.001	-0.008	0.003	-0.010	0.000
B	2	0.012	0.022	0.012	0.011	0.015	0.016	0.014	0.008	0.012	0.009	0.008	-0.001
AA	1	0.008	0.004	0.005	0.005	0.003	0.003	0.002	-0.004	-0.003	-0.002	0.004	-0.002
AA	2	-0.006	-0.023	-0.008	-0.011	-0.004	-0.016	-0.008	-0.013	0.001	-0.013	-0.003	-0.009
AB	1	-0.034	-0.054	-0.049	-0.030	-0.026	-0.051	-0.025	-0.023	-0.005	-0.055	-0.022	-0.026
AB	2	-0.001	-0.035	-0.026	-0.029	-0.021	-0.044	-0.021	-0.029	-0.013	-0.023	-0.004	-0.020
BA	1	-0.038	-0.057	-0.045	-0.046	-0.059	-0.102	-0.066	-0.067	-0.056	-0.083	-0.047	-0.045
BA	2	0.010	-0.010	-0.005	-0.008	0.015	0.001	0.007	-0.003	0.018	-0.007	0.008	-0.008
BB	1	-0.014	-0.014	-0.017	-0.020	-0.014	-0.020	-0.025	-0.023	-0.017	-0.017	-0.012	-0.013
BB	2	-0.003	-0.010	0.008	0.001	0.008	-0.011	0.006	-0.005	0.013	-0.012	0.019	0.001
BC	1	-0.017	-0.010	-0.001	0.002	0.002	-0.004	0.007	0.001	0.001	-0.003	0.007	0.002
BC	2	0.001	-0.015	-0.000	-0.002	0.012	-0.003	0.023	0.012	0.020	0.003	0.028	0.011

Table S8: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 8. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.064	0.062	0.030	0.032	0.023	0.047	0.048	0.046	0.048	0.048	0.047	0.063
Total	2	0.064	0.053	0.048	0.047	0.041	0.060	0.068	0.059	0.061	0.076	0.071	0.059
A	1	0.035	0.025	0.035	0.027	0.030	0.018	0.035	0.033	0.042	0.031	0.051	0.034
A	2	0.046	0.050	0.035	0.025	0.033	0.037	0.027	0.036	0.032	0.032	0.034	0.040
B	1	0.044	0.033	0.040	0.035	0.032	0.030	0.029	0.043	0.052	0.045	0.039	0.041
B	2	0.043	0.035	0.044	0.038	0.027	0.038	0.042	0.062	0.053	0.056	0.061	0.049
AA	1	0.038	0.033	0.041	0.023	0.028	0.039	0.042	0.039	0.036	0.040	0.048	0.045
AA	2	0.044	0.037	0.033	0.024	0.035	0.035	0.031	0.038	0.047	0.041	0.041	0.039
AB	1	0.037	0.029	0.026	0.022	0.038	0.036	0.030	0.028	0.037	0.038	0.032	0.030
AB	2	0.035	0.024	0.029	0.019	0.030	0.027	0.038	0.026	0.035	0.024	0.028	0.021
BA	1	0.028	0.037	0.032	0.016	0.030	0.022	0.031	0.025	0.027	0.020	0.022	0.022
BA	2	0.025	0.036	0.037	0.022	0.032	0.016	0.025	0.022	0.033	0.020	0.018	0.028
BB	1	0.022	0.027	0.027	0.011	0.018	0.015	0.021	0.017	0.033	0.031	0.035	0.022
BB	2	0.010	0.015	0.002	0.008	0.025	0.012	0.016	0.015	0.026	0.019	0.021	0.028
BC	1	0.023	0.020	0.027	0.023	0.039	0.021	0.024	0.019	0.022	0.022	0.029	0.018
BC	2	0.024	0.018	0.020	0.019	0.024	0.016	0.020	0.004	0.007	0.007	0.017	0.017
ETS													
Total	1	0.019	0.016	0.009	0.010	0.016	0.016	0.011	0.006	0.011	0.012	0.010	0.005
Total	2	0.018	0.014	0.006	0.006	0.007	0.005	0.003	0.001	0.006	0.006	0.007	0.003
A	1	0.006	0.005	0.005	0.007	0.007	0.005	0.003	0.004	0.005	0.007	0.001	0.004
A	2	0.004	0.006	0.005	0.007	0.010	0.009	0.005	0.005	0.008	0.012	0.011	0.018
B	1	0.015	0.011	0.010	0.007	0.005	0.009	0.013	0.012	0.011	0.012	0.010	0.004
B	2	0.014	0.017	0.013	0.008	0.007	0.008	0.009	0.007	0.008	0.009	0.008	-0.000
AA	1	0.010	0.004	0.002	0.001	0.000	0.004	0.003	0.005	0.002	0.001	-0.000	0.005
AA	2	0.008	0.003	0.000	-0.002	-0.004	-0.003	-0.004	-0.001	-0.003	-0.007	-0.007	-0.006
AB	1	0.008	0.002	0.002	0.000	0.000	0.001	0.002	0.004	-0.001	0.004	0.006	0.005
AB	2	0.006	0.003	0.001	0.001	-0.001	0.001	0.002	0.002	-0.002	-0.000	-0.001	-0.003
BA	1	0.006	0.004	0.005	0.002	-0.000	-0.001	0.002	0.001	0.001	0.003	0.003	0.006
BA	2	0.001	0.002	0.002	-0.000	-0.000	-0.000	0.001	-0.001	-0.001	-0.000	-0.002	-0.003
BB	1	0.003	-0.001	0.000	-0.004	-0.003	-0.004	-0.004	-0.004	-0.008	-0.005	-0.002	-0.000
BB	2	0.001	-0.003	-0.002	-0.001	-0.000	-0.001	0.001	0.000	-0.002	-0.002	-0.001	0.003
BC	1	0.000	0.002	-0.002	-0.000	-0.002	-0.001	-0.002	-0.003	-0.008	-0.005	-0.002	0.001
BC	2	0.004	0.003	0.002	0.001	-0.001	0.000	0.003	0.001	-0.004	0.001	0.004	0.009
VAR													
Total	1	-0.011	-0.008	-0.029	-0.008	-0.005	-0.008	-0.047	-0.047	-0.058	-0.044	-0.060	-0.031
Total	2	0.013	0.027	0.004	0.003	0.001	0.012	-0.027	-0.018	-0.037	-0.008	-0.034	-0.019
A	1	0.006	0.007	0.014	0.008	0.015	-0.001	0.004	-0.001	0.019	0.012	0.022	0.005
A	2	-0.003	0.003	-0.003	-0.003	-0.002	0.004	-0.004	-0.001	-0.007	-0.002	-0.011	-0.009
B	1	0.011	0.010	0.001	0.005	0.002	0.002	-0.010	-0.004	-0.011	-0.003	-0.015	-0.005
B	2	0.023	0.018	0.020	0.023	0.024	0.030	0.040	0.034	0.043	0.039	0.035	0.026
AA	1	-0.004	-0.001	0.004	-0.004	-0.011	-0.011	-0.002	-0.004	-0.003	-0.005	-0.001	0.001
AA	2	0.002	-0.016	-0.017	-0.018	-0.016	-0.023	-0.025	-0.024	-0.014	-0.027	-0.012	-0.023
AB	1	-0.024	-0.055	-0.046	-0.025	-0.025	-0.060	-0.043	-0.046	-0.032	-0.071	-0.033	-0.045
AB	2	-0.026	-0.044	-0.006	-0.008	0.000	-0.038	-0.007	-0.015	-0.022	-0.047	-0.025	-0.022
BA	1	-0.063	-0.076	-0.037	-0.040	-0.064	-0.081	-0.068	-0.058	-0.069	-0.091	-0.035	-0.053
BA	2	-0.005	-0.004	0.011	0.001	0.013	0.005	0.024	0.014	0.024	0.004	0.027	0.006
BB	1	-0.016	-0.018	-0.010	-0.012	-0.004	-0.009	-0.022	-0.017	-0.010	-0.017	-0.016	-0.019
BB	2	-0.015	-0.031	-0.001	-0.004	-0.008	-0.020	0.015	0.007	0.017	-0.008	0.017	0.000
BC	1	-0.003	-0.010	0.009	0.003	-0.004	-0.016	-0.003	-0.003	-0.005	-0.015	-0.001	-0.000
BC	2	-0.003	-0.008	0.007	-0.004	0.004	-0.013	0.005	-0.000	0.015	-0.002	0.030	0.009

Table S9: RelRMSE_{Base} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 9. Values in red indicate a RelRMSE_{Base} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.042	0.043	0.050	0.058	0.066	0.085	0.081	0.086	0.076	0.079	0.077	0.079
Total	2	0.046	0.052	0.051	0.061	0.068	0.069	0.073	0.081	0.073	0.077	0.068	0.077
A	1	0.036	0.040	0.035	0.029	0.037	0.042	0.044	0.042	0.044	0.052	0.052	0.044
A	2	0.038	0.045	0.038	0.044	0.050	0.049	0.046	0.041	0.044	0.052	0.046	0.052
B	1	0.037	0.051	0.050	0.059	0.063	0.081	0.083	0.077	0.072	0.078	0.077	0.076
B	2	0.032	0.037	0.041	0.042	0.041	0.043	0.048	0.049	0.041	0.051	0.051	0.059
AA	1	0.017	0.010	0.013	0.009	0.013	0.006	0.008	0.010	0.014	0.005	0.012	0.012
AA	2	0.015	0.009	0.009	0.007	0.012	0.007	0.013	0.008	0.011	0.005	0.008	0.005
AB	1	0.024	0.018	0.027	0.014	0.021	0.009	0.023	0.012	0.015	0.010	0.012	0.008
AB	2	0.010	-0.010	0.007	0.001	0.000	0.001	0.013	0.007	0.001	-0.003	0.007	-0.001
BA	1	0.018	0.014	0.009	0.005	0.010	0.001	0.005	-0.002	0.001	-0.006	-0.001	0.000
BA	2	0.016	0.007	0.008	0.003	0.009	0.002	0.006	-0.001	0.006	-0.003	0.002	-0.002
BB	1	0.016	0.011	0.015	0.010	0.017	0.008	0.006	0.008	0.010	-0.003	-0.005	-0.004
BB	2	0.015	0.006	0.006	-0.000	0.010	0.005	0.007	0.003	0.005	-0.004	-0.003	-0.009
BC	1	0.015	0.009	0.014	0.002	0.007	0.006	0.001	0.005	0.003	0.005	0.003	-0.000
BC	2	0.013	0.003	0.008	0.004	0.011	0.014	0.010	0.006	0.010	0.010	0.008	0.005
ETS													
Total	1	0.002	0.006	0.005	0.004	0.002	0.005	0.003	0.002	0.000	0.003	0.002	0.003
Total	2	0.005	0.007	0.006	0.005	0.003	0.003	0.001	0.002	0.001	0.003	0.002	0.003
A	1	0.008	0.006	0.005	0.003	0.003	0.001	0.002	0.002	0.002	0.001	0.001	0.000
A	2	0.010	0.004	0.002	0.003	0.002	0.001	0.001	0.001	-0.000	0.001	0.001	0.001
B	1	-0.003	0.003	0.002	0.003	0.001	0.002	0.001	0.001	0.000	0.000	0.001	0.001
B	2	0.006	0.005	0.002	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001	-0.000
AA	1	0.001	0.001	0.001	-0.000	0.000	0.000	0.001	0.001	0.002	0.002	0.002	0.000
AA	2	0.001	0.002	0.002	0.001	0.003	0.000	0.001	0.001	0.003	0.001	0.001	0.000
AB	1	0.007	0.003	0.004	0.001	0.002	0.001	0.001	0.001	0.002	0.000	0.000	-0.001
AB	2	0.006	0.002	0.004	0.004	0.004	0.002	0.003	0.003	0.004	0.002	0.002	0.001
BA	1	0.007	0.002	0.003	0.002	0.002	-0.001	-0.000	0.000	0.002	0.000	0.001	0.001
BA	2	0.005	0.004	0.003	0.003	0.003	0.002	0.003	0.002	0.002	0.001	0.002	0.002
BB	1	0.007	0.002	0.002	0.002	0.002	0.001	0.002	0.002	0.002	0.002	0.001	0.001
BB	2	0.001	-0.000	-0.001	-0.000	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.002
BC	1	0.005	0.001	0.001	0.002	0.003	0.002	0.001	0.002	0.002	0.001	-0.000	0.000
BC	2	0.005	0.000	-0.000	-0.000	0.000	-0.000	0.001	0.001	0.001	0.001	0.000	-0.000
VAR													
Total	1	0.003	0.022	-0.012	-0.006	-0.000	0.023	0.002	0.019	0.011	0.030	-0.001	0.003
Total	2	0.028	0.041	0.014	0.026	0.021	0.038	0.016	0.024	0.013	0.031	0.004	0.016
A	1	0.001	-0.011	-0.001	0.003	0.015	-0.007	0.013	0.008	0.014	-0.007	0.019	0.011
A	2	0.012	0.004	0.001	0.000	-0.001	0.004	-0.004	0.005	-0.001	0.004	0.002	0.007
B	1	0.005	0.007	-0.005	0.002	-0.004	0.006	-0.001	0.003	-0.009	0.006	-0.013	-0.000
B	2	0.008	0.012	0.024	0.012	0.014	0.010	0.014	0.007	0.007	0.014	0.017	0.007
AA	1	0.006	-0.001	-0.003	0.001	0.003	-0.004	-0.002	0.000	-0.001	-0.002	-0.002	0.000
AA	2	-0.031	-0.015	-0.007	-0.008	-0.019	-0.026	-0.012	-0.014	-0.004	-0.019	0.003	-0.000
AB	1	-0.061	-0.089	-0.064	-0.063	-0.077	-0.076	-0.044	-0.047	-0.042	-0.064	-0.036	-0.036
AB	2	-0.027	-0.035	-0.020	-0.028	-0.031	-0.039	-0.013	-0.033	-0.015	-0.019	-0.003	-0.031
BA	1	-0.050	-0.088	-0.072	-0.081	-0.071	-0.095	-0.080	-0.070	-0.050	-0.081	-0.045	-0.050
BA	2	-0.011	-0.005	0.004	0.008	0.015	-0.008	0.015	0.004	0.009	-0.011	0.012	-0.005
BB	1	-0.020	-0.013	-0.010	-0.013	-0.014	-0.016	-0.011	-0.016	-0.024	-0.021	-0.013	-0.012
BB	2	0.001	-0.006	0.001	-0.001	0.012	-0.015	0.001	0.002	0.023	-0.008	0.023	0.007
BC	1	-0.011	-0.008	-0.001	-0.009	-0.002	-0.007	-0.006	-0.005	0.008	0.002	0.007	-0.003
BC	2	0.005	-0.025	-0.010	-0.013	0.002	-0.013	0.003	-0.010	0.007	-0.005	0.023	-0.007

Table S10: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 1. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.003	0.005	0.007	0.005	0.004	0.005	0.002	-0.001	0.000	-0.001	-0.004	-0.008
Total	2	0.004	0.002	0.003	0.002	0.001	0.002	-0.001	-0.001	-0.001	-0.004	-0.005	-0.007
A	1	0.006	0.005	0.005	0.003	0.004	0.005	0.004	0.002	0.001	0.002	0.001	-0.002
A	2	0.005	0.002	0.003	0.002	-0.000	0.000	-0.002	0.000	0.002	-0.002	-0.003	-0.005
B	1	-0.001	0.002	0.006	0.006	0.006	0.003	0.002	0.002	0.003	-0.001	-0.002	-0.005
B	2	-0.000	-0.001	0.002	0.002	0.002	-0.000	-0.003	-0.003	-0.000	-0.004	-0.003	-0.005
AA	1	0.000	0.003	0.001	-0.001	-0.002	-0.000	-0.001	-0.002	-0.003	0.000	-0.000	-0.000
AA	2	0.005	0.003	0.004	0.004	-0.000	0.002	0.001	0.003	0.002	0.001	-0.001	0.001
AB	1	0.007	0.006	0.006	0.004	0.007	0.004	0.003	-0.000	0.003	0.002	0.001	-0.002
AB	2	-0.002	-0.000	0.000	-0.000	-0.002	-0.003	-0.004	-0.006	-0.003	-0.006	-0.007	-0.010
BA	1	0.000	0.002	0.004	0.006	0.005	0.000	0.001	0.002	0.005	0.002	0.001	0.001
BA	2	0.001	0.001	0.004	0.005	0.003	0.003	0.002	-0.001	-0.001	-0.001	0.001	-0.001
BB	1	-0.001	-0.001	0.003	0.001	0.001	0.001	-0.000	0.001	0.001	-0.001	-0.000	-0.001
BB	2	-0.002	-0.003	-0.002	-0.002	-0.001	-0.003	-0.007	-0.007	-0.004	-0.004	-0.006	-0.007
BC	1	-0.001	-0.001	0.001	0.001	0.003	-0.000	0.000	-0.001	-0.001	-0.003	-0.004	-0.005
BC	2	0.000	-0.001	0.001	0.001	0.001	0.000	0.001	-0.000	0.001	0.000	0.001	-0.000
ETS													
Total	1	0.003	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Total	2	0.003	0.002	0.001	0.001	0.001	0.000	0.000	0.000	0.001	0.001	0.001	0.001
A	1	0.002	0.000	0.000	0.000	-0.000	-0.000	0.000	0.000	-0.000	0.000	0.001	0.000
A	2	0.001	0.000	0.000	0.000	0.000	-0.001	-0.000	-0.000	0.000	-0.000	-0.000	-0.000
B	1	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
B	2	0.002	0.002	0.001	0.001	0.000	0.001	0.001	0.000	0.000	0.001	0.001	0.001
AA	1	0.001	-0.000	-0.000	-0.000	-0.000	-0.001	-0.001	-0.001	-0.000	-0.000	-0.000	-0.000
AA	2	0.001	0.001	0.000	0.000	0.000	-0.000	-0.001	-0.000	-0.000	-0.000	-0.000	-0.000
AB	1	0.001	0.000	0.000	0.000	0.000	-0.000	-0.000	0.000	0.000	-0.000	0.000	0.000
AB	2	0.000	-0.000	0.000	0.000	-0.000	-0.001	-0.001	-0.001	-0.000	-0.000	-0.000	-0.000
BA	1	0.000	0.000	0.000	0.000	0.001	-0.000	-0.000	0.000	-0.000	0.000	0.000	0.000
BA	2	0.001	0.001	0.001	0.000	0.001	0.001	0.000	0.000	0.000	-0.000	-0.000	0.000
BB	1	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
BB	2	0.001	0.001	0.000	0.000	-0.001	-0.001	-0.000	-0.000	0.000	0.000	-0.000	0.000
BC	1	0.000	0.000	0.000	0.000	-0.000	-0.000	-0.000	0.000	0.000	0.000	0.000	-0.000
BC	2	0.001	0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	0.000	0.000	0.000
VAR													
Total	1	-0.021	-0.014	-0.005	-0.013	-0.017	-0.020	-0.004	-0.011	-0.011	-0.020	-0.011	-0.023
Total	2	-0.001	-0.001	0.006	-0.004	-0.002	-0.005	-0.006	-0.009	-0.003	-0.014	0.000	-0.013
A	1	-0.015	-0.015	0.004	-0.009	-0.014	-0.014	0.005	-0.009	-0.007	-0.006	0.014	-0.009
A	2	-0.026	-0.023	-0.018	-0.016	-0.025	-0.019	-0.027	-0.016	-0.023	-0.026	-0.036	-0.017
B	1	-0.014	-0.009	-0.015	-0.010	-0.017	-0.018	-0.014	-0.004	-0.011	-0.011	-0.024	-0.016
B	2	0.003	0.002	0.005	-0.001	0.003	-0.006	-0.005	-0.007	0.009	-0.004	0.012	-0.001
AA	1	-0.007	-0.001	-0.014	-0.001	-0.001	0.001	-0.021	-0.004	-0.013	-0.002	-0.025	-0.000
AA	2	-0.019	-0.020	-0.015	-0.014	-0.031	-0.025	-0.026	-0.013	-0.020	-0.023	-0.035	-0.016
AB	1	-0.051	-0.053	-0.031	-0.027	-0.057	-0.058	-0.032	-0.044	-0.048	-0.045	-0.019	-0.037
AB	2	-0.038	-0.036	-0.027	-0.028	-0.042	-0.036	-0.044	-0.040	-0.036	-0.034	-0.043	-0.035
BA	1	-0.058	-0.055	-0.045	-0.037	-0.072	-0.092	-0.074	-0.051	-0.057	-0.069	-0.068	-0.067
BA	2	-0.005	-0.009	-0.000	-0.003	-0.002	-0.006	-0.010	-0.010	0.002	0.001	-0.001	-0.002
BB	1	-0.013	-0.014	-0.038	-0.029	-0.052	-0.035	-0.039	-0.026	-0.051	-0.030	-0.043	-0.023
BB	2	-0.005	-0.001	0.002	-0.011	-0.006	-0.005	-0.009	-0.006	0.002	0.001	0.003	-0.003
BC	1	-0.017	-0.006	-0.016	-0.008	-0.015	-0.010	-0.021	-0.014	-0.021	-0.006	-0.024	-0.009
BC	2	0.002	-0.009	-0.003	-0.004	-0.009	-0.019	-0.016	-0.015	-0.009	-0.012	-0.009	-0.008

Table S11: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 2. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.004	0.001	0.004	0.007	0.008	0.009	0.009	0.008	0.012	0.010	0.009	0.005
Total	2	-0.004	-0.007	-0.001	0.001	0.002	0.006	0.003	0.004	0.006	0.006	0.004	0.001
A	1	0.001	0.004	0.008	0.006	0.006	0.005	0.005	0.002	0.004	0.001	0.003	0.002
A	2	0.004	-0.003	0.001	0.003	0.003	0.006	0.004	0.002	0.006	0.003	0.004	0.002
B	1	0.006	-0.001	0.001	0.006	0.008	0.007	0.005	0.007	0.010	0.012	0.009	0.004
B	2	-0.002	-0.001	0.002	0.002	0.005	0.005	0.001	0.004	0.003	0.002	0.002	0.001
AA	1	-0.003	0.001	0.004	0.002	0.004	0.004	0.004	0.000	0.003	0.003	0.005	0.003
AA	2	0.001	-0.000	0.001	0.002	0.004	0.005	0.005	0.003	0.008	0.001	0.002	0.003
AB	1	-0.000	-0.000	0.006	0.004	0.003	0.000	0.003	-0.002	-0.000	-0.002	0.002	0.003
AB	2	0.003	-0.000	0.001	0.001	0.002	0.002	0.001	-0.002	0.002	0.001	0.000	-0.001
BA	1	-0.000	-0.001	-0.001	0.001	0.003	0.002	0.004	-0.002	0.001	0.004	0.004	0.002
BA	2	0.001	0.003	0.002	0.004	0.004	0.006	0.003	0.004	0.006	0.009	0.008	0.006
BB	1	-0.005	-0.004	-0.002	0.000	-0.002	-0.000	-0.001	0.003	0.005	0.005	0.005	0.002
BB	2	0.003	0.000	0.000	-0.001	0.000	0.002	0.002	0.006	0.004	0.002	0.003	0.003
BC	1	0.002	0.003	0.003	0.004	0.003	0.004	0.003	0.005	0.005	0.007	0.003	0.003
BC	2	-0.001	-0.003	0.001	0.003	0.002	0.000	-0.001	0.001	-0.001	-0.003	-0.004	-0.001
ETS													
Total	1	0.002	0.001	-0.000	-0.000	0.001	0.001	-0.000	0.000	0.001	0.002	0.002	0.001
Total	2	0.003	0.002	0.001	0.001	0.001	0.000	0.000	0.001	0.002	0.001	0.002	0.003
A	1	0.002	0.001	-0.000	-0.000	0.000	0.001	0.000	-0.000	0.000	0.002	0.002	0.001
A	2	0.001	0.001	-0.000	-0.001	-0.001	-0.002	-0.002	-0.001	0.000	0.001	0.002	0.001
B	1	0.002	0.000	0.000	0.001	0.001	0.000	0.000	0.000	0.001	0.001	0.001	0.001
B	2	0.003	0.002	0.002	0.002	0.002	0.001	0.000	0.001	0.003	0.002	0.001	0.002
AA	1	0.002	0.002	0.000	0.000	0.001	0.001	0.001	0.000	0.000	0.001	0.001	0.001
AA	2	0.000	0.000	-0.001	-0.001	-0.001	-0.001	-0.001	-0.000	0.000	0.001	0.001	0.001
AB	1	0.001	0.001	0.000	-0.001	-0.000	0.001	0.000	0.000	0.001	0.002	0.001	0.001
AB	2	0.001	0.001	-0.000	-0.001	-0.000	-0.001	-0.002	-0.001	-0.000	-0.001	0.001	0.001
BA	1	0.001	-0.000	-0.000	-0.001	-0.000	-0.001	-0.002	-0.001	-0.001	-0.000	-0.001	-0.001
BA	2	0.002	0.001	0.001	0.001	0.000	-0.000	-0.000	0.000	0.002	0.002	0.001	0.000
BB	1	0.000	-0.000	-0.000	-0.001	-0.000	-0.000	-0.001	-0.001	0.001	0.001	0.001	0.000
BB	2	0.003	0.002	0.001	0.000	0.000	-0.000	0.000	0.000	0.001	0.002	0.002	0.002
BC	1	0.001	0.000	0.000	0.000	0.002	0.002	0.001	0.000	0.001	0.001	0.001	0.001
BC	2	0.001	0.001	-0.001	-0.001	0.000	-0.000	-0.000	0.000	0.002	0.001	0.001	0.002
VAR													
Total	1	-0.034	-0.030	-0.014	-0.026	-0.040	-0.042	-0.026	-0.040	-0.041	-0.045	-0.008	-0.033
Total	2	-0.010	-0.009	-0.003	-0.012	-0.015	-0.032	-0.010	-0.016	-0.017	-0.023	0.005	-0.017
A	1	-0.041	-0.040	-0.016	-0.012	-0.011	-0.027	-0.017	-0.021	-0.007	-0.027	0.000	-0.012
A	2	-0.016	-0.014	-0.018	-0.012	-0.020	-0.027	-0.023	-0.018	-0.042	-0.028	-0.023	-0.034
B	1	-0.023	-0.008	-0.013	-0.020	-0.029	-0.025	-0.023	-0.035	-0.042	-0.028	-0.026	-0.030
B	2	-0.004	-0.019	-0.006	-0.016	-0.016	-0.032	-0.010	-0.016	-0.004	-0.007	0.024	0.002
AA	1	-0.007	-0.009	-0.013	0.002	-0.001	0.005	-0.013	-0.001	-0.013	-0.005	-0.030	-0.003
AA	2	-0.008	-0.017	-0.023	-0.019	-0.014	-0.017	-0.019	-0.020	-0.026	-0.025	-0.031	-0.031
AB	1	-0.080	-0.093	-0.064	-0.050	-0.076	-0.103	-0.095	-0.069	-0.054	-0.086	-0.065	-0.046
AB	2	-0.040	-0.043	-0.030	-0.026	-0.035	-0.049	-0.041	-0.032	-0.059	-0.039	-0.024	-0.039
BA	1	-0.086	-0.097	-0.064	-0.062	-0.110	-0.120	-0.113	-0.110	-0.110	-0.113	-0.068	-0.077
BA	2	-0.008	-0.009	-0.005	-0.010	-0.006	-0.005	-0.005	-0.005	-0.006	-0.005	0.004	-0.001
BB	1	-0.032	-0.020	-0.018	-0.015	-0.025	-0.018	-0.026	-0.009	-0.030	-0.031	-0.050	-0.030
BB	2	-0.006	-0.016	-0.009	-0.013	-0.015	-0.032	-0.024	-0.020	-0.023	-0.019	-0.001	-0.005
BC	1	-0.000	0.002	-0.009	-0.007	-0.013	0.002	-0.008	-0.004	-0.017	-0.002	-0.014	-0.005
BC	2	-0.010	-0.016	-0.009	-0.010	-0.016	-0.021	-0.017	-0.013	-0.005	-0.012	-0.002	-0.006

Table S12: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 3. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	-0.004	-0.002	-0.000	-0.002	-0.006	-0.008	-0.010	-0.009	-0.012	-0.014	-0.016	-0.016
Total	2	0.001	0.001	-0.001	-0.004	-0.004	-0.004	-0.006	-0.011	-0.011	-0.013	-0.012	-0.014
A	1	-0.001	-0.004	0.001	-0.002	-0.006	-0.007	-0.005	-0.006	-0.010	-0.010	-0.009	-0.010
A	2	0.001	0.002	0.001	-0.004	-0.004	-0.004	-0.004	-0.005	-0.007	-0.008	-0.008	-0.007
B	1	-0.002	0.002	-0.001	-0.005	-0.005	-0.009	-0.010	-0.009	-0.010	-0.013	-0.014	-0.013
B	2	0.003	0.004	-0.001	-0.002	-0.004	-0.003	-0.004	-0.008	-0.011	-0.010	-0.008	-0.010
AA	1	-0.002	-0.002	0.002	-0.002	-0.005	-0.004	-0.002	-0.002	-0.006	-0.004	-0.003	-0.004
AA	2	0.001	0.002	0.003	-0.000	-0.001	-0.002	-0.000	-0.001	-0.004	-0.004	-0.003	-0.005
AB	1	0.001	-0.001	0.002	-0.001	-0.002	-0.002	-0.003	-0.004	-0.005	-0.006	-0.006	-0.007
AB	2	0.003	0.003	-0.000	-0.006	-0.005	-0.004	-0.003	-0.004	-0.005	-0.007	-0.006	-0.005
BA	1	0.000	0.002	-0.001	-0.002	-0.006	-0.007	-0.004	-0.005	-0.007	-0.007	-0.005	-0.006
BA	2	0.003	0.005	-0.001	-0.007	-0.001	0.000	-0.001	-0.003	-0.002	-0.001	-0.001	-0.006
BB	1	-0.002	-0.002	-0.003	-0.003	-0.003	-0.007	-0.006	-0.004	-0.004	-0.008	-0.009	-0.006
BB	2	0.001	0.001	0.002	0.003	0.001	0.001	0.002	-0.001	-0.001	-0.001	0.001	-0.002
BC	1	-0.000	0.001	-0.002	-0.001	-0.002	-0.004	-0.008	-0.003	-0.002	-0.003	-0.006	-0.004
BC	2	-0.001	-0.001	-0.002	-0.002	-0.004	-0.003	-0.001	-0.001	-0.001	-0.004	-0.001	-0.002
ETS													
Total	1	0.001	0.001	0.000	0.001	0.001	0.000	-0.000	-0.000	-0.001	-0.001	-0.001	-0.001
Total	2	0.001	0.001	0.001	0.001	0.000	0.000	0.000	-0.000	-0.000	-0.000	-0.000	-0.001
A	1	0.002	0.001	0.001	0.001	0.001	0.000	0.000	0.000	0.000	-0.000	-0.000	-0.000
A	2	0.002	0.002	0.001	0.001	0.000	0.001	0.001	0.000	0.000	0.000	0.000	0.000
B	1	0.002	0.002	0.001	0.001	0.001	0.001	0.000	-0.000	-0.000	0.000	-0.000	-0.000
B	2	0.001	0.001	0.001	0.001	0.001	0.001	0.000	0.000	-0.000	0.000	-0.000	-0.000
AA	1	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.000	-0.000	-0.000	-0.000
AA	2	0.001	0.001	0.000	0.000	-0.000	-0.000	0.000	-0.000	-0.000	-0.000	-0.000	-0.000
AB	1	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.000	-0.000	-0.000	0.000	-0.000
AB	2	0.001	0.001	0.000	0.001	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000
BA	1	0.001	0.001	0.001	0.001	0.000	0.001	0.001	0.000	-0.000	-0.000	-0.000	-0.000
BA	2	0.001	0.001	0.001	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.000
BB	1	0.001	0.002	0.001	0.001	0.001	0.000	0.000	0.000	-0.000	0.000	-0.000	-0.000
BB	2	0.001	0.001	-0.000	0.001	0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
BC	1	0.001	0.001	0.001	0.000	-0.000	-0.000	-0.000	-0.000	-0.001	-0.000	-0.000	-0.000
BC	2	0.001	0.001	0.000	0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
VAR													
Total	1	-0.024	-0.031	-0.017	-0.015	-0.010	-0.032	-0.015	-0.014	0.009	-0.009	0.001	-0.007
Total	2	-0.007	-0.008	-0.004	-0.007	0.001	-0.014	-0.000	-0.004	0.011	-0.005	0.009	-0.001
A	1	-0.022	-0.010	-0.005	-0.002	-0.006	-0.011	0.002	-0.006	0.004	-0.008	0.006	-0.004
A	2	-0.016	-0.013	-0.014	-0.021	-0.019	-0.014	-0.007	-0.001	-0.006	-0.011	-0.013	0.005
B	1	-0.019	-0.026	-0.019	-0.013	-0.011	-0.023	-0.028	-0.018	0.003	-0.004	-0.015	-0.011
B	2	-0.006	-0.009	0.004	0.001	0.006	-0.009	0.002	-0.004	0.010	-0.012	0.005	-0.006
AA	1	-0.003	0.009	0.001	-0.006	-0.004	0.003	-0.020	-0.010	-0.021	0.002	-0.023	-0.004
AA	2	-0.015	-0.011	-0.016	-0.019	-0.017	-0.014	-0.008	-0.013	-0.012	-0.008	-0.013	-0.003
AB	1	-0.044	-0.056	-0.038	-0.029	-0.051	-0.060	-0.032	-0.032	-0.036	-0.042	-0.029	-0.027
AB	2	-0.027	-0.025	-0.015	-0.030	-0.034	-0.031	-0.022	-0.008	-0.019	-0.028	-0.021	-0.009
BA	1	-0.047	-0.085	-0.054	-0.046	-0.069	-0.088	-0.074	-0.059	-0.052	-0.061	-0.045	-0.046
BA	2	-0.013	-0.003	0.006	-0.004	-0.001	-0.007	-0.003	-0.001	0.002	-0.002	0.002	-0.006
BB	1	-0.015	-0.020	-0.032	-0.026	-0.030	-0.028	-0.038	-0.021	-0.028	-0.022	-0.034	-0.025
BB	2	-0.000	-0.001	-0.001	-0.002	-0.002	-0.014	-0.006	-0.008	-0.002	-0.010	-0.005	-0.004
BC	1	-0.007	-0.001	-0.009	-0.010	-0.010	-0.006	-0.025	-0.011	-0.005	0.005	-0.020	-0.006
BC	2	-0.004	-0.009	-0.000	-0.004	-0.001	-0.010	-0.011	-0.010	0.002	-0.009	-0.007	-0.010

Table S13: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 4. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.006	0.009	0.009	0.006	-0.001	0.003	0.003	-0.000	0.001	0.001	0.000	-0.003
Total	2	0.004	0.006	0.008	0.004	-0.002	0.001	0.000	-0.004	-0.004	-0.003	-0.003	-0.003
A	1	0.010	0.013	0.005	-0.001	-0.001	0.003	0.001	-0.001	-0.002	-0.001	-0.003	-0.005
A	2	0.004	0.010	0.007	-0.001	-0.003	0.001	-0.001	-0.004	-0.003	-0.004	-0.004	-0.004
B	1	0.003	0.007	0.007	0.009	0.004	0.006	0.002	0.000	-0.000	0.000	-0.001	-0.005
B	2	0.006	0.003	0.007	0.006	0.003	-0.000	-0.001	-0.003	-0.000	-0.002	-0.003	-0.006
AA	1	0.011	0.007	0.006	0.001	0.002	0.002	0.000	0.001	0.004	0.003	0.000	-0.002
AA	2	0.011	0.011	0.004	0.000	0.002	0.001	-0.001	-0.002	-0.003	-0.006	-0.004	-0.005
AB	1	0.004	0.009	0.003	0.002	0.003	0.007	0.004	-0.000	0.000	-0.002	-0.003	-0.006
AB	2	0.004	0.007	0.004	0.003	0.001	-0.000	-0.002	-0.004	0.001	-0.002	-0.007	-0.005
BA	1	0.003	0.006	0.004	0.003	0.005	0.000	0.001	-0.002	-0.003	0.001	0.000	-0.001
BA	2	0.003	0.006	0.006	0.004	0.003	0.000	0.001	-0.002	-0.001	0.001	-0.002	0.001
BB	1	0.003	0.003	0.006	0.007	0.004	0.006	0.003	0.003	0.000	0.002	-0.000	-0.000
BB	2	0.004	0.004	0.009	0.007	0.007	0.004	-0.001	0.001	-0.000	-0.000	-0.004	-0.005
BC	1	0.005	0.005	0.008	0.010	0.008	0.004	0.000	-0.000	0.002	-0.002	-0.003	-0.001
BC	2	0.005	0.004	0.004	0.000	0.002	-0.000	-0.002	-0.003	0.001	0.002	0.001	0.001
ETS													
Total	1	0.005	0.003	0.002	0.001	0.000	0.000	0.000	0.000	-0.000	0.000	0.001	0.000
Total	2	0.004	0.002	0.001	0.001	-0.001	-0.001	-0.002	-0.001	-0.001	-0.001	-0.001	-0.000
A	1	0.003	0.002	0.000	-0.000	0.001	0.001	0.000	0.000	0.000	0.000	0.000	-0.000
A	2	0.002	0.001	-0.000	-0.001	-0.000	-0.000	-0.001	-0.001	-0.000	-0.000	-0.001	-0.000
B	1	0.004	0.003	0.002	0.001	0.000	0.001	-0.000	-0.000	-0.001	-0.000	-0.000	-0.000
B	2	0.004	0.002	0.002	0.002	-0.000	-0.000	-0.000	0.000	-0.000	0.000	-0.000	-0.000
AA	1	0.003	0.001	0.001	0.001	0.002	0.002	0.002	0.001	0.000	0.001	0.001	0.000
AA	2	0.002	0.001	0.000	-0.001	0.000	-0.000	-0.001	-0.000	0.000	0.000	-0.000	-0.001
AB	1	0.002	0.003	0.002	0.001	0.001	0.001	0.001	0.001	0.002	0.002	0.001	0.001
AB	2	0.001	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.000	0.000
BA	1	0.002	0.002	0.001	0.001	0.001	0.000	-0.001	0.000	-0.000	0.000	-0.000	0.000
BA	2	0.003	0.001	0.002	0.001	0.000	-0.000	0.000	-0.000	-0.000	0.000	0.000	-0.000
BB	1	0.003	0.003	0.002	0.001	0.001	0.001	0.001	0.000	-0.000	-0.000	0.000	0.000
BB	2	0.001	0.001	0.001	0.001	-0.001	-0.001	-0.001	-0.001	-0.001	0.000	0.000	0.000
BC	1	0.003	0.002	0.001	0.001	0.000	0.001	0.000	-0.000	-0.000	-0.000	0.000	0.000
BC	2	0.002	0.002	0.002	0.001	0.001	0.001	0.000	0.001	0.001	0.000	0.000	0.000
VAR													
Total	1	-0.042	-0.054	-0.041	-0.045	-0.027	-0.039	-0.034	-0.035	-0.022	-0.036	-0.021	-0.035
Total	2	-0.028	-0.038	-0.032	-0.032	-0.030	-0.031	-0.013	-0.020	-0.013	-0.020	-0.001	-0.026
A	1	-0.032	-0.034	-0.011	-0.028	-0.017	-0.026	-0.028	-0.021	-0.008	-0.030	-0.021	-0.031
A	2	-0.028	-0.034	-0.032	-0.024	-0.034	-0.040	-0.044	-0.038	-0.033	-0.031	-0.028	-0.027
B	1	-0.029	-0.044	-0.049	-0.038	-0.030	-0.038	-0.038	-0.032	-0.037	-0.029	-0.030	-0.026
B	2	-0.026	-0.026	-0.012	-0.029	-0.024	-0.026	-0.009	-0.023	-0.013	-0.021	0.003	-0.015
AA	1	-0.017	-0.005	-0.009	-0.005	-0.026	-0.011	-0.029	-0.022	-0.044	-0.026	-0.025	-0.005
AA	2	-0.031	-0.031	-0.029	-0.030	-0.040	-0.023	-0.026	-0.028	-0.032	-0.011	-0.019	-0.011
AB	1	-0.078	-0.070	-0.060	-0.070	-0.070	-0.080	-0.091	-0.059	-0.050	-0.075	-0.085	-0.074
AB	2	-0.038	-0.055	-0.038	-0.022	-0.026	-0.037	-0.044	-0.036	-0.024	-0.046	-0.043	-0.044
BA	1	-0.089	-0.091	-0.080	-0.087	-0.114	-0.136	-0.122	-0.094	-0.100	-0.092	-0.074	-0.091
BA	2	-0.023	-0.021	-0.026	-0.029	-0.027	-0.022	-0.009	-0.012	-0.015	-0.016	-0.009	-0.014
BB	1	-0.029	-0.028	-0.033	-0.022	-0.029	-0.025	-0.037	-0.020	-0.039	-0.033	-0.036	-0.023
BB	2	-0.026	-0.023	-0.013	-0.022	-0.016	-0.022	-0.016	-0.024	-0.020	-0.020	-0.011	-0.014
BC	1	-0.014	-0.016	-0.027	-0.011	-0.014	-0.010	-0.026	-0.018	-0.026	-0.011	-0.030	-0.014
BC	2	-0.031	-0.018	-0.009	-0.017	-0.013	-0.022	-0.008	-0.012	-0.010	-0.010	-0.001	-0.007

Table S14: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 5. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.006	0.004	0.002	-0.005	-0.004	-0.001	-0.002	-0.001	0.000	0.002	0.001	0.001
Total	2	0.010	0.008	0.010	0.002	0.001	0.002	-0.000	0.003	0.002	0.001	-0.000	-0.001
A	1	-0.000	0.003	0.001	0.000	0.001	0.000	-0.000	-0.000	0.000	-0.001	-0.001	-0.002
A	2	0.005	0.006	0.006	0.005	0.003	0.005	-0.001	-0.000	0.001	-0.001	-0.001	0.000
B	1	0.007	0.004	-0.000	-0.004	-0.005	0.000	0.000	0.002	0.001	0.004	0.002	0.005
B	2	0.006	0.006	0.006	0.000	0.003	0.003	0.000	0.002	0.001	0.003	0.003	0.001
AA	1	0.002	0.005	0.003	0.001	0.000	0.003	-0.001	-0.000	-0.001	-0.003	-0.002	-0.001
AA	2	0.003	0.005	0.001	-0.000	0.001	0.001	-0.001	-0.002	0.002	-0.002	-0.000	0.004
AB	1	0.002	0.002	0.001	0.002	-0.001	0.000	-0.001	-0.003	-0.002	-0.001	0.003	0.004
AB	2	0.005	0.007	0.005	0.008	-0.000	0.001	-0.004	-0.003	0.001	-0.003	-0.005	-0.001
BA	1	0.007	0.004	0.005	0.004	-0.001	0.001	0.001	-0.000	0.001	0.005	0.002	0.007
BA	2	-0.001	0.005	-0.001	-0.001	-0.001	0.003	0.005	0.005	0.002	0.002	0.001	0.006
BB	1	0.005	0.000	0.002	0.001	-0.001	-0.001	0.005	0.006	0.002	0.004	0.004	0.006
BB	2	0.008	0.001	0.002	0.001	0.002	0.002	0.001	0.002	-0.001	0.001	0.001	0.000
BC	1	0.007	0.007	0.008	0.001	-0.001	0.000	-0.001	-0.002	-0.004	-0.004	-0.000	-0.002
BC	2	0.002	0.003	0.003	0.007	0.006	0.002	-0.001	-0.002	-0.000	0.001	-0.000	0.001
ETS													
Total	1	0.005	0.002	0.004	0.003	0.002	0.001	-0.000	-0.001	0.001	0.001	-0.001	-0.001
Total	2	0.005	0.003	0.002	0.001	0.001	-0.001	-0.001	-0.000	0.003	0.002	0.003	0.002
A	1	0.006	0.004	0.003	0.002	0.001	0.001	0.001	0.001	0.002	0.002	0.001	-0.000
A	2	0.002	0.002	0.003	0.002	0.001	0.001	0.000	-0.000	0.003	0.001	-0.000	0.001
B	1	0.004	0.003	0.003	0.003	0.001	-0.000	0.002	0.001	0.001	0.001	0.002	0.000
B	2	0.006	0.004	0.004	0.002	0.002	0.000	0.001	0.002	0.004	0.004	0.005	0.003
AA	1	0.000	-0.001	-0.000	0.002	0.002	0.002	0.002	0.001	0.001	0.002	0.001	-0.001
AA	2	0.002	0.002	0.002	0.003	0.001	0.001	0.002	-0.000	0.002	0.003	0.001	0.001
AB	1	0.009	0.005	0.004	0.003	0.001	0.001	0.001	-0.000	0.000	-0.001	-0.001	-0.001
AB	2	0.002	0.003	0.002	-0.001	0.001	-0.000	-0.000	-0.001	0.002	0.001	-0.001	0.000
BA	1	0.004	0.004	0.003	0.001	0.001	0.002	0.004	0.000	0.002	0.003	0.003	0.001
BA	2	0.005	0.004	0.005	0.002	0.002	0.002	0.002	0.002	0.002	0.004	0.003	0.002
BB	1	0.006	0.002	0.001	0.002	0.001	0.001	0.002	0.002	0.002	0.002	0.001	0.001
BB	2	0.007	0.002	0.001	0.002	-0.000	-0.000	-0.000	0.000	-0.001	0.001	0.000	0.001
BC	1	0.006	0.005	0.004	0.003	0.001	0.002	0.001	0.001	0.002	0.001	-0.001	-0.000
BC	2	0.003	0.002	0.003	0.002	0.001	0.000	0.001	0.001	0.002	0.002	0.001	0.001
VAR													
Total	1	-0.040	-0.056	-0.053	-0.060	-0.057	-0.066	-0.045	-0.051	-0.056	-0.057	-0.039	-0.059
Total	2	-0.016	-0.033	-0.019	-0.039	-0.038	-0.041	-0.026	-0.039	-0.023	-0.037	-0.017	-0.022
A	1	-0.048	-0.059	-0.028	-0.043	-0.044	-0.041	-0.027	-0.020	-0.018	-0.026	-0.009	-0.023
A	2	-0.031	-0.043	-0.031	-0.026	-0.022	-0.027	-0.033	-0.029	-0.040	-0.043	-0.050	-0.039
B	1	-0.033	-0.031	-0.036	-0.035	-0.047	-0.051	-0.042	-0.048	-0.058	-0.048	-0.051	-0.044
B	2	-0.020	-0.023	-0.011	-0.027	-0.026	-0.036	-0.032	-0.027	-0.013	-0.038	-0.013	-0.018
AA	1	-0.021	-0.013	-0.017	-0.006	-0.010	-0.005	-0.012	-0.002	-0.030	-0.023	-0.056	-0.026
AA	2	-0.021	-0.033	-0.041	-0.037	-0.042	-0.048	-0.048	-0.035	-0.047	-0.049	-0.056	-0.049
AB	1	-0.091	-0.108	-0.091	-0.099	-0.101	-0.131	-0.100	-0.069	-0.082	-0.090	-0.095	-0.073
AB	2	-0.043	-0.072	-0.049	-0.027	-0.033	-0.064	-0.058	-0.042	-0.051	-0.071	-0.062	-0.047
BA	1	-0.095	-0.125	-0.127	-0.118	-0.155	-0.154	-0.140	-0.129	-0.161	-0.197	-0.176	-0.145
BA	2	-0.020	-0.031	-0.026	-0.025	-0.031	-0.030	-0.032	-0.018	-0.012	-0.011	0.001	-0.011
BB	1	-0.044	-0.029	-0.024	-0.027	-0.042	-0.039	-0.063	-0.046	-0.053	-0.039	-0.052	-0.028
BB	2	-0.012	-0.027	-0.021	-0.015	-0.022	-0.027	-0.027	-0.009	-0.007	-0.024	-0.015	-0.011
BC	1	-0.026	-0.018	-0.016	-0.017	-0.027	-0.031	-0.034	-0.028	-0.027	-0.006	-0.024	-0.006
BC	2	-0.018	-0.017	-0.010	-0.030	-0.030	-0.030	-0.015	-0.024	-0.018	-0.023	-0.004	-0.013

Table S15: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 6. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.011	0.012	0.011	0.002	0.003	-0.000	0.000	-0.001	-0.003	-0.005	-0.006	-0.008
Total	2	0.016	0.012	0.011	0.003	0.008	0.001	0.002	0.002	0.003	-0.002	-0.002	-0.007
A	1	0.007	0.013	0.007	0.003	0.005	0.000	-0.001	0.001	-0.003	-0.009	-0.008	-0.007
A	2	0.010	0.013	0.014	0.010	0.011	0.006	0.006	0.003	0.002	0.001	0.003	-0.002
B	1	0.017	0.009	0.011	0.004	0.008	0.005	0.005	0.004	0.003	0.000	-0.001	-0.000
B	2	0.011	0.003	0.004	-0.001	0.002	-0.003	-0.003	-0.004	-0.004	-0.006	-0.004	-0.008
AA	1	0.006	0.013	0.004	-0.004	0.003	0.003	0.000	-0.001	0.000	-0.003	-0.005	-0.008
AA	2	0.006	0.009	0.010	0.009	0.012	0.005	0.002	0.003	0.003	0.001	0.001	-0.000
AB	1	0.015	0.019	0.016	0.008	0.014	0.008	0.004	0.002	0.002	-0.001	-0.002	-0.002
AB	2	0.009	0.011	0.011	0.006	0.008	0.005	0.003	0.001	0.001	-0.002	0.001	-0.003
BA	1	0.017	0.010	0.014	0.005	0.008	0.006	0.007	0.003	0.005	0.003	0.006	0.003
BA	2	0.006	0.002	0.005	0.002	0.003	-0.000	0.004	0.004	0.002	0.000	0.008	0.006
BB	1	0.007	0.005	0.009	0.004	0.004	-0.003	0.002	0.002	0.003	-0.001	0.001	0.000
BB	2	0.013	0.007	0.001	-0.000	0.004	0.004	0.000	-0.004	-0.002	-0.001	-0.004	-0.010
BC	1	0.008	0.004	0.004	0.005	0.007	0.009	0.007	0.002	0.002	0.004	0.003	0.000
BC	2	0.009	0.003	0.005	0.001	0.004	0.004	0.005	0.002	0.002	0.001	0.001	-0.004
ETS													
Total	1	0.006	0.005	0.008	0.007	0.006	0.004	0.004	0.004	0.004	0.004	0.004	0.003
Total	2	0.006	0.006	0.007	0.004	0.004	0.002	0.002	0.002	0.002	0.003	0.002	0.002
A	1	0.006	0.005	0.005	0.004	0.004	0.003	0.002	0.003	0.003	0.003	0.002	0.003
A	2	0.004	0.004	0.004	0.002	0.003	0.003	0.002	0.002	0.002	0.002	0.001	0.001
B	1	0.006	0.005	0.006	0.005	0.005	0.003	0.003	0.002	0.002	0.002	0.002	0.001
B	2	0.006	0.005	0.006	0.005	0.005	0.001	0.001	0.001	0.001	0.000	-0.000	0.001
AA	1	0.006	0.005	0.003	0.001	0.002	0.002	0.001	0.002	0.001	0.002	0.001	0.001
AA	2	0.003	0.002	0.002	0.000	0.001	0.002	0.001	0.000	0.001	0.001	0.000	0.000
AB	1	0.004	0.004	0.004	0.003	0.003	0.002	0.001	0.002	0.002	0.002	0.001	0.001
AB	2	0.004	0.004	0.003	0.001	0.001	0.002	0.001	0.001	0.001	0.001	0.000	-0.000
BA	1	0.003	0.003	0.004	0.003	0.003	0.002	0.002	0.002	0.002	0.001	0.002	0.002
BA	2	0.004	0.003	0.004	0.003	0.003	0.002	0.002	0.002	0.002	0.001	0.001	0.001
BB	1	0.005	0.004	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.000	0.000
BB	2	0.003	0.001	0.002	0.000	0.000	-0.000	0.001	-0.001	-0.001	-0.001	-0.001	-0.001
BC	1	0.003	0.001	0.003	0.001	0.001	0.000	0.001	0.000	0.000	-0.000	0.000	-0.000
BC	2	0.004	0.004	0.003	0.002	0.002	0.001	-0.000	-0.000	-0.000	0.000	0.000	-0.000
VAR													
Total	1	-0.052	-0.071	-0.045	-0.031	-0.024	-0.057	-0.044	-0.033	-0.007	-0.025	-0.016	-0.027
Total	2	-0.039	-0.045	-0.038	-0.038	-0.026	-0.036	-0.019	-0.024	-0.005	-0.017	-0.005	-0.014
A	1	-0.033	-0.037	-0.034	-0.022	-0.008	-0.028	-0.021	-0.025	-0.008	-0.015	-0.005	-0.012
A	2	-0.043	-0.042	-0.027	-0.033	-0.052	-0.050	-0.043	-0.033	-0.038	-0.020	-0.013	-0.014
B	1	-0.044	-0.059	-0.040	-0.033	-0.038	-0.059	-0.046	-0.033	-0.025	-0.028	-0.031	-0.029
B	2	-0.031	-0.046	-0.033	-0.030	-0.018	-0.030	-0.005	-0.011	-0.001	-0.011	0.003	-0.012
AA	1	-0.018	-0.026	-0.030	-0.013	-0.019	-0.021	-0.037	-0.021	-0.028	-0.003	-0.020	-0.011
AA	2	-0.035	-0.035	-0.024	-0.029	-0.032	-0.021	-0.035	-0.021	-0.017	-0.016	-0.021	-0.006
AB	1	-0.070	-0.091	-0.082	-0.044	-0.050	-0.072	-0.075	-0.060	-0.053	-0.055	-0.052	-0.041
AB	2	-0.055	-0.069	-0.030	-0.033	-0.063	-0.060	-0.036	-0.029	-0.047	-0.034	-0.025	-0.036
BA	1	-0.097	-0.115	-0.099	-0.082	-0.103	-0.140	-0.099	-0.079	-0.067	-0.081	-0.075	-0.069
BA	2	-0.029	-0.037	-0.033	-0.029	-0.023	-0.021	-0.010	-0.010	-0.007	-0.011	-0.005	-0.006
BB	1	-0.034	-0.036	-0.041	-0.030	-0.046	-0.040	-0.046	-0.038	-0.036	-0.007	-0.033	-0.026
BB	2	-0.013	-0.030	-0.037	-0.030	-0.022	-0.031	-0.022	-0.020	-0.014	-0.018	-0.008	-0.012
BC	1	-0.026	-0.028	-0.018	-0.010	-0.015	-0.010	-0.014	-0.008	-0.025	-0.007	-0.018	-0.007
BC	2	-0.026	-0.025	-0.029	-0.013	-0.013	-0.018	-0.005	-0.004	-0.002	-0.008	-0.006	-0.005

Table S16: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 7. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.005	0.002	0.004	0.000	0.000	-0.005	0.001	-0.002	-0.000	-0.003	-0.002	-0.004
Total	2	0.005	0.003	0.001	-0.001	0.002	-0.001	0.000	0.000	0.001	0.001	0.000	-0.002
A	1	0.004	0.008	0.005	0.003	0.004	0.002	0.005	0.003	0.001	0.002	0.001	0.001
A	2	0.004	0.002	0.003	0.001	0.003	0.004	0.002	0.002	0.001	0.003	0.001	0.001
B	1	0.004	0.000	0.002	-0.001	0.000	-0.003	-0.001	-0.002	0.000	-0.002	-0.000	-0.005
B	2	0.004	0.001	0.000	0.002	0.005	0.001	-0.001	0.002	0.001	0.001	-0.002	-0.003
AA	1	0.003	0.007	0.002	0.004	0.004	0.002	0.003	0.001	0.002	0.002	0.001	0.002
AA	2	0.003	0.002	0.004	0.001	0.005	0.004	0.002	0.002	0.003	0.002	0.001	0.001
AB	1	-0.000	0.001	0.003	0.001	0.004	-0.000	0.002	0.003	0.001	0.000	0.001	0.001
AB	2	-0.000	0.002	0.001	0.002	0.001	0.003	0.002	0.001	-0.000	0.003	0.001	-0.002
BA	1	0.003	-0.003	-0.001	0.000	0.002	-0.003	-0.002	-0.001	0.001	-0.003	-0.001	-0.002
BA	2	0.002	0.000	0.002	0.003	0.002	0.000	0.001	0.003	0.001	0.002	0.001	0.001
BB	1	0.005	0.002	-0.000	-0.001	0.001	-0.001	0.002	0.001	0.001	0.003	0.001	-0.002
BB	2	0.001	0.000	-0.001	-0.002	-0.000	-0.002	-0.001	0.001	0.002	0.000	-0.000	-0.003
BC	1	-0.000	0.003	0.002	-0.001	0.001	0.000	0.001	0.000	0.002	0.000	0.000	-0.004
BC	2	0.001	0.004	-0.000	0.000	0.001	0.002	0.000	0.002	0.001	0.002	0.001	-0.000
ETS													
Total	1	0.002	0.001	0.000	0.001	0.000	-0.000	-0.000	0.000	0.000	0.000	0.000	-0.000
Total	2	0.002	0.003	0.001	0.001	0.000	0.000	0.001	0.001	0.001	0.001	0.001	-0.000
A	1	0.001	0.001	0.001	0.001	0.001	0.000	0.001	0.000	0.000	0.000	0.000	-0.000
A	2	0.000	0.001	0.001	0.001	0.000	0.000	0.001	0.000	0.001	0.001	0.000	-0.000
B	1	0.001	0.000	0.000	-0.000	-0.000	-0.000	0.000	0.000	0.000	0.000	0.000	0.000
B	2	0.001	0.002	0.001	0.001	0.000	-0.000	0.000	0.000	0.000	0.000	0.000	-0.000
AA	1	0.001	0.001	0.001	0.001	-0.000	0.000	0.000	0.000	0.001	0.001	0.000	0.000
AA	2	0.001	0.001	0.001	0.000	-0.000	0.000	0.000	0.000	0.001	0.000	-0.000	-0.000
AB	1	0.001	0.001	0.001	0.001	-0.000	-0.000	0.000	0.000	0.000	0.000	0.000	-0.000
AB	2	0.000	-0.000	0.000	-0.000	-0.000	-0.001	-0.000	-0.000	-0.000	-0.000	-0.000	-0.001
BA	1	0.001	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0.001	0.000	0.000	-0.000
BA	2	0.001	0.003	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.000
BB	1	0.001	0.001	0.000	0.001	0.001	0.000	0.001	0.000	0.001	0.001	0.001	0.000
BB	2	0.001	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.000
BC	1	0.002	0.002	0.001	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.000	0.001
BC	2	0.001	0.002	0.001	0.000	0.001	0.001	0.001	0.000	0.001	0.000	-0.000	-0.000
VAR													
Total	1	-0.021	-0.033	-0.006	-0.011	-0.014	-0.030	-0.007	-0.017	-0.013	-0.026	-0.000	-0.012
Total	2	-0.006	-0.019	-0.006	-0.012	-0.000	-0.006	0.002	-0.010	0.000	-0.007	0.006	-0.003
A	1	-0.017	-0.017	-0.004	-0.006	0.002	-0.006	0.008	-0.003	0.004	-0.013	0.003	-0.008
A	2	-0.014	-0.021	-0.012	-0.009	-0.020	-0.025	-0.019	-0.021	-0.026	-0.020	-0.020	-0.016
B	1	-0.022	-0.022	-0.009	-0.012	-0.023	-0.032	-0.019	-0.015	-0.023	-0.017	-0.007	-0.003
B	2	-0.007	-0.015	-0.005	-0.007	0.006	-0.004	0.006	-0.003	0.008	-0.006	0.011	0.000
AA	1	-0.008	-0.004	-0.017	-0.005	-0.012	-0.002	-0.017	-0.008	-0.024	-0.005	-0.018	-0.001
AA	2	-0.020	-0.030	-0.026	-0.016	-0.014	-0.016	-0.023	-0.019	-0.027	-0.024	-0.024	-0.008
AB	1	-0.047	-0.052	-0.051	-0.038	-0.041	-0.055	-0.036	-0.029	-0.032	-0.052	-0.038	-0.028
AB	2	-0.015	-0.036	-0.027	-0.024	-0.033	-0.043	-0.031	-0.033	-0.038	-0.029	-0.025	-0.022
BA	1	-0.050	-0.058	-0.056	-0.054	-0.072	-0.099	-0.080	-0.072	-0.080	-0.082	-0.060	-0.041
BA	2	-0.003	-0.011	-0.012	-0.005	0.001	-0.001	-0.001	-0.003	-0.001	-0.001	-0.005	-0.007
BB	1	-0.023	-0.016	-0.026	-0.025	-0.025	-0.012	-0.030	-0.018	-0.030	-0.011	-0.025	-0.013
BB	2	-0.008	-0.011	0.002	-0.003	-0.002	-0.006	0.000	-0.002	0.000	-0.004	0.001	0.003
BC	1	-0.020	-0.014	-0.015	-0.011	-0.013	-0.013	-0.016	-0.006	-0.016	-0.008	-0.019	-0.005
BC	2	-0.009	-0.013	-0.010	-0.007	-0.002	-0.008	0.000	0.003	-0.000	-0.002	0.003	0.002

Table S17: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 8. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.004	0.003	0.004	0.003	0.004	0.004	0.001	0.002	0.003	0.004	0.004	0.003
Total	2	0.006	0.005	0.004	0.005	0.005	0.004	-0.002	-0.003	-0.001	0.001	0.003	0.007
A	1	0.011	0.005	0.006	0.003	0.007	0.004	0.005	0.006	0.005	0.008	0.006	0.002
A	2	0.007	0.007	0.004	0.004	0.004	0.002	-0.000	-0.000	0.004	0.003	0.004	0.005
B	1	0.001	0.004	0.005	0.004	0.006	0.006	0.004	0.007	0.006	0.008	0.009	0.008
B	2	-0.002	0.001	0.001	0.005	0.004	0.005	0.003	0.002	0.003	0.003	0.002	0.006
AA	1	0.006	0.004	0.004	0.001	0.004	0.004	0.003	0.006	0.005	0.010	0.007	0.007
AA	2	0.005	0.005	0.004	0.001	0.003	0.005	0.007	0.010	0.008	0.010	0.010	0.008
AB	1	0.010	0.006	0.006	0.002	0.007	0.002	0.004	0.007	0.009	0.009	0.007	0.004
AB	2	0.006	0.006	0.002	0.005	0.002	0.001	0.002	-0.002	0.001	-0.000	-0.000	0.001
BA	1	0.001	0.004	0.002	0.003	0.003	0.000	0.004	0.006	0.004	0.007	0.004	0.003
BA	2	0.001	0.004	0.002	0.001	0.007	0.004	-0.000	0.001	0.004	0.001	0.002	0.003
BB	1	-0.002	0.001	0.001	-0.000	-0.001	-0.001	0.000	0.003	0.002	0.002	0.003	0.003
BB	2	-0.003	0.001	0.001	0.007	0.004	0.005	0.004	0.005	0.005	0.004	-0.001	0.001
BC	1	0.000	0.001	0.003	0.001	0.004	0.003	0.001	0.005	0.006	0.007	0.003	0.003
BC	2	0.003	0.001	0.000	0.000	0.005	0.004	0.004	0.001	0.004	0.003	0.002	0.004
ETS													
Total	1	0.005	0.003	0.002	0.002	0.000	-0.000	0.001	0.002	0.002	0.004	0.002	0.002
Total	2	0.004	0.003	0.001	0.001	-0.000	-0.002	-0.001	0.001	0.002	0.002	0.001	0.003
A	1	0.005	0.002	0.002	0.001	0.002	0.002	0.002	0.002	0.002	0.003	0.002	0.002
A	2	0.004	0.003	0.001	0.001	0.003	0.002	-0.000	0.000	0.001	0.001	0.001	0.004
B	1	0.003	0.004	0.002	0.003	0.001	-0.000	-0.000	0.002	0.001	0.003	0.002	0.002
B	2	0.004	0.004	0.001	0.002	0.001	-0.001	-0.000	0.001	0.001	0.001	0.000	0.001
AA	1	0.004	0.002	0.002	0.002	0.001	0.002	0.002	0.001	0.002	0.002	0.002	0.000
AA	2	0.003	0.002	-0.000	0.001	0.001	0.000	-0.000	-0.000	0.001	0.000	0.000	0.001
AB	1	0.003	0.002	0.001	-0.000	0.001	0.001	-0.000	0.001	0.002	0.003	0.003	0.002
AB	2	0.002	0.002	-0.000	0.001	0.001	0.001	-0.000	0.001	0.001	0.002	0.002	0.004
BA	1	0.002	0.003	0.002	0.001	0.001	-0.000	0.001	0.001	0.001	0.002	0.001	0.001
BA	2	0.001	0.002	0.002	0.000	0.001	0.001	0.001	0.000	0.000	0.001	-0.001	0.000
BB	1	0.003	0.001	0.002	0.001	-0.001	-0.000	0.001	0.001	-0.000	0.002	0.002	0.001
BB	2	0.003	0.002	-0.001	0.000	0.000	0.000	0.000	0.001	0.001	0.001	0.001	-0.001
BC	1	0.002	0.002	0.001	0.001	0.000	-0.001	-0.001	0.001	0.000	0.001	0.001	0.000
BC	2	0.003	0.003	0.003	0.002	0.002	0.000	0.002	0.002	0.001	0.001	0.002	0.001
VAR													
Total	1	-0.021	-0.044	-0.036	-0.020	-0.017	-0.041	-0.034	-0.039	-0.028	-0.047	-0.020	-0.029
Total	2	-0.008	-0.019	-0.011	-0.015	0.000	-0.013	-0.004	-0.009	-0.004	-0.017	-0.000	-0.012
A	1	-0.009	-0.020	-0.015	-0.017	-0.007	-0.029	-0.016	-0.025	-0.010	-0.026	-0.005	-0.016
A	2	-0.011	-0.023	-0.026	-0.020	-0.017	-0.022	-0.025	-0.023	-0.028	-0.027	-0.034	-0.027
B	1	-0.015	-0.035	-0.027	-0.011	-0.022	-0.032	-0.039	-0.035	-0.042	-0.044	-0.032	-0.026
B	2	-0.010	-0.017	0.005	-0.001	0.000	-0.008	0.014	0.007	0.017	0.001	0.018	0.007
AA	1	-0.021	-0.010	-0.008	-0.008	-0.020	-0.002	-0.006	-0.006	-0.018	0.002	-0.015	-0.009
AA	2	-0.019	-0.022	-0.027	-0.024	-0.026	-0.027	-0.034	-0.027	-0.036	-0.028	-0.034	-0.031
AB	1	-0.035	-0.063	-0.060	-0.039	-0.044	-0.064	-0.056	-0.051	-0.056	-0.074	-0.053	-0.050
AB	2	-0.035	-0.049	-0.018	-0.014	-0.018	-0.043	-0.027	-0.026	-0.044	-0.047	-0.043	-0.023
BA	1	-0.069	-0.086	-0.062	-0.055	-0.081	-0.094	-0.094	-0.072	-0.087	-0.097	-0.066	-0.063
BA	2	-0.008	-0.012	-0.007	-0.007	0.005	0.007	0.004	0.004	0.010	0.004	-0.000	-0.002
BB	1	-0.019	-0.011	-0.016	-0.013	-0.004	0.000	-0.031	-0.024	-0.023	-0.020	-0.034	-0.022
BB	2	-0.009	-0.016	-0.004	-0.009	-0.017	-0.017	-0.003	-0.005	-0.006	-0.015	-0.004	-0.006
BC	1	-0.009	-0.011	-0.003	0.002	-0.012	-0.011	-0.021	-0.017	-0.024	-0.019	-0.023	-0.009
BC	2	-0.010	-0.005	-0.002	-0.005	-0.011	-0.009	-0.004	-0.006	-0.003	-0.006	0.005	0.001

Table S18: RelRMSE_{Uni} for all series in the hierarchy across different forecast horizons. ARIMA, ETS, and VAR models for base forecasts. Using the shrinkage approach to estimate $\mathbf{W}$ under scenario 9. Values in red indicate a RelRMSE_{Uni} less than 0.

Series	Variable	Forecast horizons (h)											
		1	2	3	4	5	6	7	8	9	10	11	12
ARIMA													
Total	1	0.002	0.004	0.002	-0.001	-0.001	-0.001	-0.004	-0.009	-0.006	-0.007	-0.009	-0.012
Total	2	-0.003	-0.005	-0.002	-0.003	-0.005	-0.006	-0.003	-0.005	-0.008	-0.010	-0.008	-0.008
A	1	0.003	0.004	0.003	0.002	0.000	-0.001	-0.003	-0.003	-0.003	-0.004	-0.006	-0.006
A	2	0.001	0.000	0.001	-0.001	0.002	-0.001	0.001	-0.003	-0.006	-0.005	-0.005	-0.006
B	1	0.002	0.005	0.002	0.000	0.001	0.001	-0.003	-0.006	-0.005	-0.006	-0.008	-0.010
B	2	-0.000	-0.003	0.001	-0.000	0.001	-0.004	-0.005	-0.005	-0.005	-0.008	-0.009	-0.008
AA	1	0.002	0.002	0.003	0.002	0.001	-0.001	-0.002	-0.001	-0.002	-0.003	-0.003	-0.003
AA	2	0.004	0.007	0.004	0.000	0.004	0.001	0.003	-0.001	-0.002	-0.001	-0.003	-0.004
AB	1	0.000	0.004	0.003	0.002	0.001	0.000	-0.002	-0.001	-0.003	-0.002	-0.003	-0.004
AB	2	-0.001	-0.003	-0.002	-0.003	0.001	-0.001	-0.001	-0.004	-0.003	-0.007	-0.006	-0.008
BA	1	0.002	0.003	0.002	0.002	0.002	0.003	0.002	-0.001	-0.001	-0.002	-0.000	-0.002
BA	2	0.003	0.001	-0.001	-0.001	-0.000	-0.002	-0.000	-0.005	-0.003	-0.004	-0.004	-0.003
BB	1	0.001	0.003	0.003	0.002	0.002	0.003	0.003	0.000	0.001	-0.001	-0.003	-0.004
BB	2	-0.003	-0.002	0.001	0.001	0.003	0.002	0.001	0.000	0.000	-0.000	-0.001	-0.000
BC	1	0.006	0.002	0.003	-0.001	0.001	-0.000	-0.001	-0.001	-0.000	-0.002	-0.003	-0.006
BC	2	0.001	-0.003	0.001	-0.001	-0.001	-0.003	-0.003	-0.002	-0.003	-0.004	-0.003	-0.002
ETS													
Total	1	0.001	0.003	0.002	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Total	2	0.001	0.001	0.001	0.000	0.000	0.000	0.000	0.001	0.001	0.000	-0.000	-0.000
A	1	0.002	0.003	0.002	0.001	0.001	0.000	0.001	0.001	0.001	0.001	0.000	0.000
A	2	0.002	0.002	0.003	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.000
B	1	-0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
B	2	0.001	0.001	0.001	0.000	0.001	0.001	0.000	0.000	0.000	0.000	-0.000	-0.000
AA	1	0.001	0.001	0.001	0.001	0.001	0.000	0.001	0.001	0.001	0.000	0.001	0.000
AA	2	0.001	0.002	0.002	0.001	0.001	0.000	0.001	0.001	0.001	0.000	0.000	0.000
AB	1	-0.000	0.001	0.001	0.000	0.001	0.000	0.000	0.000	0.000	0.000	-0.000	-0.000
AB	2	0.000	0.000	0.001	0.001	0.001	-0.000	0.000	0.001	0.001	0.000	0.000	0.000
BA	1	-0.001	0.001	0.002	0.002	0.002	0.001	0.001	0.001	0.001	0.001	0.001	0.001
BA	2	0.002	0.001	0.001	0.001	0.001	0.000	0.000	0.000	0.001	0.000	-0.000	0.000
BB	1	0.001	0.000	-0.000	0.001	0.000	0.000	0.001	0.000	0.000	0.000	-0.000	0.000
BB	2	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
BC	1	0.001	0.002	0.000	0.001	0.001	0.002	0.002	0.001	0.001	0.001	0.001	0.001
BC	2	0.001	0.001	0.001	0.000	0.000	0.001	0.000	0.000	0.000	0.000	-0.000	-0.000
VAR													
Total	1	-0.035	-0.026	-0.022	-0.030	-0.024	-0.038	-0.016	-0.007	-0.004	-0.017	0.009	-0.009
Total	2	-0.015	-0.021	-0.017	-0.008	-0.003	-0.017	-0.005	0.002	0.007	-0.007	0.012	0.002
A	1	-0.035	-0.031	-0.020	-0.018	-0.016	-0.033	-0.011	-0.011	-0.005	-0.018	0.001	-0.002
A	2	-0.032	-0.022	-0.017	-0.015	-0.035	-0.028	-0.023	-0.018	-0.020	-0.009	-0.015	-0.008
B	1	-0.019	-0.017	-0.027	-0.025	-0.020	-0.024	-0.026	-0.014	-0.012	-0.016	-0.014	-0.010
B	2	-0.010	-0.016	-0.008	-0.006	0.003	-0.017	-0.004	-0.000	0.009	-0.009	0.017	0.000
AA	1	0.006	0.006	-0.006	0.001	-0.005	0.003	-0.004	0.002	-0.015	0.004	-0.014	0.001
AA	2	-0.025	-0.010	-0.013	-0.009	-0.022	-0.026	-0.030	-0.010	-0.014	-0.010	-0.010	0.001
AB	1	-0.074	-0.092	-0.074	-0.064	-0.082	-0.081	-0.060	-0.051	-0.052	-0.058	-0.046	-0.036
AB	2	-0.033	-0.037	-0.025	-0.028	-0.039	-0.038	-0.025	-0.034	-0.026	-0.015	-0.020	-0.028
BA	1	-0.059	-0.099	-0.085	-0.082	-0.086	-0.101	-0.091	-0.069	-0.067	-0.074	-0.057	-0.049
BA	2	-0.013	-0.010	-0.003	0.003	0.004	-0.002	0.006	0.000	-0.004	-0.007	0.001	-0.006
BB	1	-0.029	-0.023	-0.028	-0.020	-0.033	-0.017	-0.029	-0.026	-0.046	-0.019	-0.034	-0.017
BB	2	-0.008	-0.003	-0.009	-0.008	-0.006	-0.013	-0.011	-0.005	0.002	-0.009	0.005	0.004
BC	1	-0.016	-0.004	-0.020	-0.013	-0.011	0.001	-0.009	0.002	-0.008	-0.002	-0.018	-0.002
BC	2	-0.004	-0.019	-0.020	-0.017	-0.011	-0.015	-0.011	-0.013	-0.006	-0.012	-0.004	-0.010

Table S19: RelRMSE^{Base} of admission series. VAR model for base forecasts and the shrinkage approach to estimate $\mathbf{W}$. Values in red indicate a RelRMSE^{Base} less than 0.

Series	Forecast horizons (h)											
	1	2	3	4	5	6	7	8	9	10	11	12
Total	-0.020	-0.089	-0.049	-0.080	-0.118	-0.176	-0.147	-0.087	-0.062	-0.124	-0.212	-0.402
Midwest	-0.045	-0.065	-0.035	-0.039	-0.083	-0.107	-0.091	-0.079	-0.063	-0.172	-0.178	-0.235
Northeast	0.033	-0.047	-0.029	-0.109	-0.119	-0.160	-0.160	-0.145	-0.141	-0.173	-0.245	-0.363
North	0.019	-0.029	-0.028	-0.031	-0.070	-0.081	-0.052	-0.042	-0.023	-0.141	-0.142	-0.123
Southeast	-0.003	-0.065	-0.026	-0.044	-0.102	-0.169	-0.153	-0.131	-0.094	-0.171	-0.249	-0.403
South	-0.033	-0.067	0.003	0.014	-0.003	-0.061	-0.060	-0.043	-0.035	-0.089	-0.172	-0.290
AC	-0.019	-0.021	-0.017	-0.021	-0.020	-0.024	-0.022	-0.020	-0.025	-0.037	-0.063	-0.140
AL	0.115	0.094	0.100	0.141	0.154	0.195	0.203	0.220	0.190	0.159	0.149	0.225
AM	-0.034	-0.077	-0.086	-0.119	-0.157	-0.211	-0.216	-0.205	-0.177	-0.210	-0.287	-0.306
AP	-0.016	-0.034	-0.038	-0.052	-0.057	-0.064	-0.070	-0.082	-0.091	-0.107	-0.113	-0.099
BA	-0.009	-0.023	-0.026	-0.033	-0.064	-0.077	-0.077	-0.060	-0.050	-0.059	-0.067	-0.073
CE	0.021	-0.009	0.000	-0.042	-0.065	-0.077	-0.060	-0.028	-0.006	-0.029	-0.063	-0.149
DF	0.012	0.002	-0.007	-0.029	-0.056	-0.098	-0.066	-0.036	-0.037	-0.059	-0.139	-0.165
ES	-0.012	-0.049	-0.036	-0.030	-0.030	-0.034	-0.036	-0.035	-0.030	-0.031	-0.027	-0.033
GO	-0.027	-0.053	-0.031	-0.029	-0.046	-0.056	-0.067	-0.057	-0.067	-0.166	-0.206	-0.256
MA	-0.011	-0.046	-0.039	-0.043	-0.052	-0.059	-0.057	-0.068	-0.075	-0.130	-0.115	-0.105
MG	-0.021	-0.040	-0.020	-0.026	-0.061	-0.092	-0.096	-0.097	-0.080	-0.106	-0.115	-0.137
MS	-0.045	-0.090	-0.058	-0.072	-0.082	-0.121	-0.120	-0.123	-0.132	-0.178	-0.201	-0.206
MT	0.005	-0.016	-0.013	-0.024	-0.033	-0.028	-0.017	-0.018	-0.035	-0.129	-0.109	-0.092
PA	0.001	-0.037	-0.030	-0.073	-0.136	-0.174	-0.159	-0.123	-0.056	-0.051	-0.067	-0.107
PB	-0.028	-0.007	-0.020	-0.023	-0.012	-0.046	-0.030	0.017	-0.020	-0.002	0.001	0.026
PE	0.019	0.011	0.009	-0.016	-0.042	-0.065	-0.069	-0.036	-0.011	-0.005	-0.017	-0.053
PI	-0.021	-0.040	-0.042	-0.048	-0.051	-0.055	-0.043	-0.032	-0.035	-0.056	-0.079	-0.080
PR	-0.036	-0.059	-0.012	-0.003	-0.015	-0.062	-0.047	-0.033	-0.052	-0.111	-0.233	-0.362
RJ	0.015	-0.033	-0.028	-0.047	-0.066	-0.091	-0.087	-0.070	-0.079	-0.089	-0.107	-0.122
RN	0.004	-0.000	-0.018	-0.022	-0.028	-0.032	-0.024	-0.013	-0.024	-0.031	-0.038	-0.034
RO	-0.017	-0.017	-0.016	-0.037	-0.065	-0.069	-0.058	-0.044	-0.044	-0.058	-0.071	-0.083
RR	0.013	-0.040	-0.022	-0.056	-0.065	-0.078	-0.064	-0.037	-0.038	-0.024	-0.054	-0.048
RS	-0.060	-0.109	-0.032	-0.013	-0.043	-0.127	-0.149	-0.148	-0.115	-0.217	-0.281	-0.286
SC	-0.024	-0.022	0.040	0.052	0.035	0.006	0.027	0.036	0.030	-0.011	-0.032	-0.131
SE	-0.011	-0.018	-0.004	-0.048	-0.060	-0.113	-0.097	-0.065	-0.035	-0.051	-0.091	-0.161
SP	-0.043	-0.116	-0.053	-0.038	-0.028	-0.074	-0.068	-0.034	-0.037	-0.072	-0.142	-0.386
TO	-0.008	-0.030	-0.015	-0.034	-0.058	-0.070	-0.055	-0.035	-0.026	-0.058	-0.080	-0.119

Table S20: RelRMSE^{Base} of dismissal series. VAR model for base forecasts and the shrinkage approach to estimate $\mathbf{W}$. Values in red indicate a RelRMSE^{Base} less than 0.

Series	Forecast horizons (h)											
	1	2	3	4	5	6	7	8	9	10	11	12
Total	0.098	0.104	0.093	0.061	0.048	0.036	0.026	0.014	-0.004	-0.054	-0.073	-0.109
Midwest	0.031	0.014	-0.022	-0.021	-0.048	-0.023	-0.046	-0.043	-0.032	-0.048	-0.043	-0.064
Northeast	0.007	0.033	0.050	0.001	0.004	-0.192	-0.303	-0.405	-0.468	-0.437	-0.448	-0.427
North	0.047	0.040	0.043	0.037	-0.022	0.047	0.043	0.022	0.039	0.054	0.092	0.045
Southeast	0.023	0.019	0.007	0.027	0.006	0.020	0.022	-0.009	0.005	-0.026	-0.023	-0.065
South	0.000	0.007	-0.005	-0.016	-0.009	-0.037	-0.038	-0.038	-0.061	-0.085	-0.117	-0.186
AC	-0.018	-0.012	-0.012	-0.013	-0.012	-0.019	-0.008	-0.018	-0.016	-0.027	-0.037	-0.037
AL	0.020	0.019	-0.011	-0.045	-0.016	-0.006	-0.007	-0.007	-0.002	0.010	0.017	0.013
AM	-0.055	-0.036	-0.012	0.095	0.145	0.077	0.078	0.094	0.140	0.156	0.098	0.144
AP	-0.027	0.044	0.051	0.038	0.038	0.021	0.036	0.014	0.012	0.008	-0.006	0.009
BA	0.020	0.037	0.037	0.065	0.062	0.065	0.083	0.069	0.067	0.061	0.060	0.052
CE	0.043	-0.008	0.010	0.012	-0.007	0.004	0.014	0.004	0.006	-0.065	-0.030	-0.041
DF	-0.001	-0.024	-0.036	0.020	0.005	0.032	0.027	0.012	0.001	0.003	0.042	0.010
ES	-0.002	0.004	0.018	0.036	0.035	0.040	0.044	0.042	0.049	0.056	0.060	0.058
GO	-0.009	0.013	-0.006	-0.055	0.008	-0.009	0.008	0.008	0.009	0.035	0.023	0.069
MA	-0.010	-0.026	-0.030	0.018	0.004	0.010	0.023	0.001	0.023	0.016	0.020	0.039
MG	0.039	0.038	0.034	0.038	0.033	0.049	0.054	0.056	0.056	0.062	0.066	0.072
MS	-0.007	-0.005	-0.002	-0.020	-0.012	-0.028	-0.025	-0.018	-0.024	-0.030	-0.035	-0.022
MT	-0.011	-0.010	-0.005	-0.037	-0.010	-0.073	-0.059	-0.058	-0.080	-0.070	-0.074	-0.063
PA	-0.009	-0.000	-0.009	0.014	0.013	0.009	0.013	0.003	0.009	-0.003	-0.014	0.005
PB	0.054	0.081	0.026	0.016	0.027	-0.016	-0.087	-0.129	-0.143	-0.156	-0.158	-0.125
PE	-0.044	-0.018	-0.004	0.007	0.026	-0.061	-0.114	-0.187	-0.184	-0.203	-0.203	-0.241
PI	-0.012	-0.011	0.018	0.048	0.022	0.056	0.062	0.060	0.055	0.038	0.026	0.048
PR	-0.013	-0.008	-0.010	-0.007	-0.004	-0.018	-0.019	-0.027	-0.038	-0.074	-0.092	-0.126
RJ	0.008	-0.002	0.008	0.034	0.027	0.029	0.027	0.025	0.037	0.038	0.046	0.027
RN	-0.012	-0.021	-0.025	0.004	0.014	-0.003	-0.007	-0.014	-0.019	-0.004	0.015	0.004
RO	0.006	-0.001	-0.005	-0.019	-0.026	-0.022	-0.016	-0.015	-0.018	-0.028	-0.034	-0.044
RR	0.021	0.010	0.002	0.031	0.034	0.034	0.053	0.030	0.015	0.016	0.023	0.063
RS	-0.000	-0.015	-0.008	-0.018	-0.015	-0.049	-0.064	-0.065	-0.078	-0.100	-0.099	-0.128
SC	-0.015	0.016	0.003	0.015	0.024	-0.021	-0.036	-0.043	-0.075	-0.107	-0.142	-0.207
SE	-0.043	-0.045	0.047	0.002	-0.013	-0.043	-0.069	-0.139	-0.139	-0.176	-0.215	-0.197
SP	-0.040	-0.049	-0.053	-0.085	-0.080	-0.068	-0.084	-0.065	-0.052	-0.024	-0.002	0.031
TO	-0.024	-0.067	-0.046	-0.048	-0.028	-0.023	-0.017	-0.008	0.024	0.007	0.006	0.033